DUALITY IN TENSOR-TRIANGULAR GEOMETRY
LECTURE NOTES

Overview. Tensor-triangular geometry provides a framework for studying classification problems across a range of areas. Originating in chromatic homotopy theory, its influence has since been seen in commutative algebra, geometry, and representation theory. In the course, we will see how one can prove powerful duality statements from a tensor-triangular perspective. We will focus on examples in commutative algebra, but the theory developed is rather broad, so other examples may be covered depending on the audience's interests.

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Relevant literature: main references.

(1) P. Balmer, I. Dell'Ambrogio, and B. Sanders. Grothendieck-Neeman duality and the Wirthmüller isomorphism. *Compos. Math.*, 152(8):1740–1776, 2016
(2) H. Fausk, P. Hu, and J. P. May. Isomorphisms between left and right adjoints. *Theory Appl. Categ.*, 11:No. 4, 107–131, 2003
(3) T. Peirce, J. Williamson. Duality in tensor-triangular geometry via proxy-smallness. arXiv:2510.24415, 2025

Related literature: other helpful references.

(1) H. Krause. Localization theory for triangulated categories. In Triangulated categories, volume 375 of London Math. Soc. Lecture Note Ser., pages 161–235. Cambridge Univ. Press, Cambridge, 2010
(2) H. Krause. Homological theory of representations, volume 195 of Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 2022
(3) M. Hovey, J. H. Palmieri, and N. P. Strickland. Axiomatic stable homotopy theory. Mem. Amer. Math. Soc., 128(610):x+114, 1997
(4) C. Weibel. An introduction to homological algebra. Cambridge Studies in Advanced Mathematics. 38. Cambridge: Cambridge University Press. xiv, 450 p. (1994).
(5) A. Neeman. Triangulated categories. Annals of Mathematics Studies. 148. Princeton, NJ: Princeton University Press. vii, 449 p. (2001).

Assessment. The final exam will be an oral exam. For zápočet, students are expected to participate during the course.

1. Recollections on triangulated categories

1.A. **Axioms.** Loosely speaking, a triangulated category consists of an additive category T, together with two extra pieces of data:

(1) an equivalence of categories $\Sigma\colon \mathsf{T} \xrightarrow{\sim} \mathsf{T}$ called the *shift*;
(2) a collection of *triangles* $X \to Y \to Z \to \Sigma X$ satisfying various axioms which ensure good behaviour.

Definition 1.1. Let T be an additive category and $\Sigma\colon \mathsf{T} \xrightarrow{\sim} \mathsf{T}$ be an additive equivalence of categories. A *candidate triangle* is a diagram

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$$

such that the composites $g \circ f$, $h \circ g$, and $\Sigma f \circ h$ are all zero. A morphism of candidate triangles is a commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u"] & Y \arrow[r, "g"] \arrow[d, "v"] & Z \arrow[r, "h"] \arrow[d, "w"] & \Sigma X \arrow[d, "\Sigma u"] \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & \Sigma X'. \end{tikzcd}$$

Considering a smaller class of candidate triangles which satisfy certain properties leads to the notion of a triangulated category. The first four axioms are easy to justify, but the final axiom is harder to motivate. It is convenient to develop the theory assuming only these first four axioms, and then add in the final one once it becomes relevant. Nonetheless, we'll give both definitions now, so that we can consider an example before embarking on the abstract theory.

Definition 1.2. A *pretriangulated category* T is an additive category together with an additive equivalence of categories $\Sigma\colon \mathsf{T} \xrightarrow{\sim} \mathsf{T}$, and a subclass of candidate triangles called *distinguished triangles* which satisfy the following axioms:

(TR0) Any candidate triangle which is isomorphic to a distinguished triangle is a distinguished triangle, and for all $X \in \mathsf{T}$ the candidate triangle

$$X \xrightarrow{1} X \to 0 \to \Sigma X$$

is distinguished.

(TR1) For all $f\colon X \to Y$ in T, there exists a distinguished triangle $X \xrightarrow{f} Y \to Z \to \Sigma X$.

(TR2) Let $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$ be a candidate triangle. This is distinguished if and only if the candidate triangle

$$Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X \xrightarrow{-\Sigma f} \Sigma Y$$

is distinguished.

(TR3) For any commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u"] & Y \arrow[r, "g"] \arrow[d, "v"] & Z \arrow[r, "h"] & \Sigma X \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & \Sigma X'. \end{tikzcd}$$

in which the rows are triangles, there exists a map $w\colon Z \to Z'$ (which need *not* be unique) making

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u"] & Y \arrow[r, "g"] \arrow[d, "v"] & Z \arrow[r, "h"] \arrow[d, "w"] & \Sigma X \arrow[d, "\Sigma u"] \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & \Sigma X'. \end{tikzcd}$$

commute.

Remark 1.3. It is standard to drop the adjective 'distinguished', and just refer to them as *triangles*. For candidate triangles, we will never drop the adjective.

Remark 1.4. By combining (TR2) with (TR3), one sees that there always exists fillers in the first and second column too.

Definition 1.5. A *triangulated category* T is a pretriangulated category satisfying the following additional axiom:

(TR4) Suppose that $X \xrightarrow{f} Y \xrightarrow{g} Z' \xrightarrow{h} \Sigma X$, $Y \xrightarrow{u} Z \xrightarrow{v} X' \xrightarrow{w} \Sigma Y$ and $X \xrightarrow{uf} Z \xrightarrow{a} Y' \xrightarrow{b} \Sigma X$ are distinguished triangles. Then there exists a distinguished triangle

$$Z' \xrightarrow{\alpha} Y' \xrightarrow{\beta} X' \xrightarrow{\gamma} \Sigma Z'$$

such that $v = \beta a$, $h = b\alpha$, $\gamma = \Sigma g \circ w$, $w\beta = \Sigma f \circ b$ and $\alpha g = au$.

Pictorially this axiom can be represented by the following commuting diagram:

$$\begin{tikzcd} & Y' \arrow[dr, dashed, "b"] & & \\ Z' \arrow[ur, "\alpha"] \arrow[d, dashed, "h"'] & & X' \arrow[ll, dashed, "\Sigma g \circ w"] \arrow[ul, "\beta"'] \arrow[dl, dashed] & \\ X \arrow[rr, "uf"] \arrow[dr, "f"'] & & Z \arrow[ul, "a"'] \arrow[u, "v"'] & \\ & Y \arrow[ul, "g"] \arrow[ur, "u"'] \arrow[uur, "w"'] & & \end{tikzcd}$$

The dotted maps are of degree 1 (i.e., $f\colon X \dashrightarrow Y$ represents a map $f\colon X \to \Sigma Y$), and composites of the form $\to\to\dashrightarrow$ are triangles. The red maps are the extra data, together with the condition that they form a triangle. In order to remember this, note that the primed letters are the cones of maps, and that every triangle contains an X, Y, and a Z (primed or otherwise). In light of the shape of the above diagram, (TR4) is often referred to as the *octahedral axiom*.

Alternatively, one can give the following pictorial representation.

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "1"] & Y \arrow[r, "g"] \arrow[d, "u"] & Z' \arrow[r, "h"] \arrow[d, dashed, "\alpha"] & \Sigma X \arrow[d, "1"] \\
X \arrow[r, "uf"] \arrow[d, "f"] & Z \arrow[r, "a"] \arrow[d, "1"] & Y' \arrow[r, "b"] \arrow[d, dashed, "\beta"] & \Sigma X \arrow[d, "\Sigma f"] \\
Y \arrow[r, "u"] \arrow[d, "g"] & Z \arrow[r, "v"] \arrow[d, "a"] & X' \arrow[r, "w"] \arrow[d, "1"] & \Sigma Y \arrow[d, "\Sigma g"] \\
Z' \arrow[r, dashed, "\alpha"] & Y' \arrow[r, dashed, "\beta"] & X' \arrow[r, dashed, "\gamma"] & \Sigma Z'
\end{tikzcd}$$

The first three rows are the given triangles, and (TR4) then asserts the existence of the dotted arrows making the diagram commute, so that the bottom row is also a triangle.

Let's briefly discuss the axioms and provide some motivation for them. If one thinks as triangles as a generalisation of short exact sequences, then in a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$ you should think of Z as the (homotopy coherent) cokernel of f, and X as the (homotopy coherent) kernel of g. The axioms then mean the following.

(TR0) The kernel and cokernel of the identity is zero.
(TR1) Every map has a kernel and cokernel.
(TR2) Up to sign, every map is the kernel of its cokernel and vice versa.
(TR3) Kernels and cokernels are almost functorial.
(TR4) One can interpret the given triangles as saying $Z' \simeq Y/X$, $X' \simeq Z/Y$ and $Y' \simeq Z/X$, and then the axiom asserts that $X' \simeq Y'/Z'$, i.e., $(Z/X)/(Y/X) \simeq Z/Y$.

Remark 1.6. Given a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \to \Sigma X$, it is common to call Z the *cofibre* (or *cone*) of f, and X the *fibre* (or *cocone*) of g. Sometimes it is customary to write triangles as $X \to Y \to Z$ and drop the map to the shift. We will sometimes subscribe to this later on the course for brevity, but we warn the reader that it is important not to forget this map. For example, a morphism of triangles requires the square including the map $Z \to \Sigma X$ to commute.

Let us recall some key features:

(1) Products and coproducts of triangles are again triangles (provided the required (co)products exist).
(2) If $X \to Y \to Z \to \Sigma X$ is a triangle, then $Z \simeq 0$ if and only if $X \to Y$ is an isomorphism.
(3) Given a commutative diagram of pretriangles

$$\begin{tikzcd}
X \arrow[r] \arrow[d, "f"] & Y \arrow[r] \arrow[d, "g"] & Z \arrow[r] \arrow[d, "h"] & \Sigma X \arrow[d, "\Sigma f"] \\
X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & \Sigma X'
\end{tikzcd}$$

if f and g are isomorphisms then h is also an isomorphism. Note that this gives a good way to show that a pretriangle is infact a triangle, since triangles are closed under isomorphism.

1.B. **Examples.** We briefly describe three examples of triangulated categories.

1.B.1. *The homotopy category of a ring.* Fix a ring R. Recall that a map $f\colon M \to N$ of chain complexes of (left) R-modules is null homotopic if it is chain homotopic to the zero map. We write $\mathrm{Null}(M, N)$ for the subgroup of $\mathrm{Hom}_{\mathrm{Ch}(R)}(M, N)$ consisting of the null homotopic maps.

The *homotopy category* $\mathsf{K}(R)$ is defined by having objects the chain complexes of R-modules, and morphisms given by the homotopy classes of chain maps, i.e.,

$$\mathrm{Hom}_{\mathsf{K}(R)}(M,N) = \mathrm{Hom}_{\mathrm{Ch}(R)}(M,N)/\mathrm{Null}(M,N).$$

Equivalently, the morphisms are the chain maps up to the equivalence relation of chain homotopy equivalence. The distinguished triangles in the homotopy category are the triangles which are isomorphic in $\mathsf{K}(R)$ (i.e., chain homotopic) to those of the form

$$M \xrightarrow{f} N \to C(f) \to \Sigma M$$

for some map of chain complexes $f\colon M \to N$.

1.B.2. *The derived category of a ring.* Homological algebra is built upon taking resolutions of modules. Therefore, one seeks a category which contains precisely the homological information of modules, so that objects are resolutions, and a module is isomorphic to any resolution of it. As such, we want a category which contains all the resolutions of modules, and in which quasi-isomorphisms are isomorphisms. Since injective resolutions and projective resolutions point in opposite directions, we consider all chain complexes (i.e., rather than just those bounded above or below 0). The *derived category* of a ring R is the universal category in which quasi-isomorphisms of complexes are inverted. We denote this category by $\mathsf{D}(R) := \mathsf{K}(R)[\text{quasi isos}^{-1}]$. We now give a precise construction of this.

Remark 1.7. This construction of the derived category leads to some set-theoretic discussions, namely, why are the hom sets actually sets? For the purposes of this course we ignore this, and just remark that one can give alternative constructions bypassing this issue.

The objects of $\mathsf{D}(R)$ are the same as the objects of $\mathsf{K}(R)$, that is, they are the chain complexes of R-modules. The morphisms in $\mathsf{D}(R)$ are equivalence classes of rooves, defined as follows. Let $M, N \in \mathsf{D}(R)$. A *roof* from M to N is a pair of chain maps

$$\begin{tikzcd} & Z \arrow[dl, "s", "\simeq"'] \arrow[dr, "f"] & \\ M & & N \end{tikzcd}$$

where $Z \in \mathsf{D}(R)$ and s is a quasi-isomorphism. Two rooves $(M \leftarrow Z \to N)$ and $(M \leftarrow Z' \to N)$ are equivalent if there exists another roof $(M \leftarrow W \to N)$ and maps $W \to Z$ and $W \to Z'$ such that the diagram

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ M & W \arrow[l] \arrow[r] \arrow[u] \arrow[d] & N \\ & Z' \arrow[ul] \arrow[ur] & \end{tikzcd}$$

commutes. The hom sets of $\mathsf{D}(R)$ are rooves up to this equivalence relation.

Remark 1.8. Defining composites requires a little work: given two rooves $(L \xleftarrow{s} Z \xrightarrow{f} M)$ and $(M \xleftarrow{s'} Z' \xrightarrow{g} N)$, their composite gf is defined as follows. We have a triangle

$$Z' \xrightarrow{s'} M \xrightarrow{\alpha} C$$

and another triangle

$$F \xrightarrow{\beta} Z \xrightarrow{\alpha f} C.$$

By the existence of fillers we have a diagram

$$\begin{tikzcd} F \arrow[r, "\beta"] \arrow[d, dashed, "\gamma'"] & Z \arrow[r, "\alpha f"] \arrow[d, "f"] & C \arrow[r] \arrow[d, "\mathrm{id}"] & \Sigma F \arrow[d, dashed, "\Sigma\beta"] \\ Z' \arrow[r, "s'"'] & M \arrow[r, "\alpha"] & C \arrow[r] & \Sigma Z' \end{tikzcd}$$

Since s' is a quasi-isomorphism, C is acyclic so β is also a quasi-isomorphism. The composite is then given by the roof

$$L \xleftarrow{s\beta} F \xrightarrow{g\gamma} N.$$

A tedious check shows everything is well-defined up to equivalence of rooves.

We end by recalling some key features of the derived category:

(1) If $0 \to L \xrightarrow{f} M \xrightarrow{g} N \to 0$ is short exact sequence of R-modules, then there exists a chain map $N \to \Sigma L$ such that $L \xrightarrow{f} M \xrightarrow{g} N \to \Sigma L$ is a triangle in $\mathsf{D}(R)$.
(2) In fact, one can generalise the previous to see that short exact sequences of complexes also give triangles in the derived category.
(3) Let M and N be R-modules. Then
$$\mathrm{Hom}_{\mathsf{D}(R)}(M, \Sigma^i N) = \mathrm{Ext}^i_R(M, N).$$
(4) For any $M \in \mathsf{D}(R)$ we have $\mathrm{Hom}_{\mathsf{D}(R)}(\Sigma^i R, M) = H_i M$.

1.B.3. *The stable module category.* Let G be a finite group and let k be a field of characteristic dividing the order of G. In this setting, Maschke's theorem fails and we are in the realm of *modular* representation theory. A key category of interest in this area is the *stable module category* $\mathsf{StMod}(kG)$.

Lemma 1.9. *A kG-module M is projective if and only if it is injective.*

Proof. We first show that kG is self-injective. Define a k-linear map $\pi\colon kG \to k$ by $\pi(g) = 1$ if $g = 1$ and $\pi(g) = 0$ otherwise. One can check that the map $kG \to \mathrm{Hom}_k(kG, k)$ defined by $g \mapsto [h \mapsto \pi(g^{-1}h)]$ is an isomorphism of kG-modules. Now

$$\mathrm{Hom}_{kG}(-, kG) = \mathrm{Hom}_{kG}(-, \mathrm{Hom}_k(kG, k)) = \mathrm{Hom}_k(\mathrm{res}^{kG}_k(-), k)$$

which is exact, hence kG is an injective kG-module.

Since kG is Noetherian, direct sums of injective modules are injective. Therefore free modules are injectives, and hence summands of free modules (i.e., projectives) are also injective. Conversely, suppose that M is injective. The unit of the induction-restriction adjunction $M \to kG \otimes_k \mathrm{res}^{kG}_k M$ is injective and hence splits. Now $kG \otimes_k \mathrm{res}^{kG}_k M$ is a free module, so M is projective. □

The objects of $\mathsf{StMod}(kG)$ are the kG-modules, and the morphisms are given by the kG-module maps modulo those which factor through some projective:

$$\mathrm{Hom}_{\mathsf{StMod}(kG)}(M, N) = \frac{\mathrm{Hom}_{kG}(M, N)}{P\mathrm{Hom}_{kG}(M, N)}$$

where $P\mathrm{Hom}_{kG}(M, N)$ denotes the linear subspace of homomorphisms which factor through some projective kG-module.

The shift functor is given by taking cosyzygies: given X we define ΣX by taking an exact sequence

$$0 \to X \to E \to \Sigma X \to 0$$

where E is injective. To show that this is an autoequivalence, one constructs Σ^{-1} by taking syzygies; here is where Lemma 1.9 comes in. Any short exact sequence $0 \to X \to Y \to Z \to 0$ fits into a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r, "\alpha"] \arrow[d] & Y \arrow[r, "\beta"] \arrow[d] & Z \arrow[r] \arrow[d, "\gamma"] & 0 \\ 0 \arrow[r] & X \arrow[r] & E \arrow[r] & \Sigma X \arrow[r] & 0 \end{tikzcd}$$

where E is injective. The triangles are those isomorphic to ones of the form

$$X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \xrightarrow{\gamma} \Sigma X$$

as in the above diagram.

We end by recalling some key features of the stable module category:

(1) We have $M \cong N$ in $\mathsf{StMod}(kG)$ if and only if there exists projectives P and Q such that $M \oplus P \cong N \oplus Q$ in $\mathsf{Mod}(kG)$.
(2) The homomorphisms in the stable module category compute Ext-groups: if $i \geq 0$ then
$$\mathrm{Hom}_{\mathsf{StMod}(kG)}(M, \Sigma^i N) = \mathrm{Ext}^i_{kG}(M, N).$$
In particular, taking $M = k$ with trivial G-action, it computes group cohomology: $H^i(G; N) = \mathrm{Hom}_{\mathsf{StMod}(kG)}(k, \Sigma^i N)$. (In fact, allowing negative i too, one obtains Tate cohomology groups.) Note that here unlike in the derived category, we are representing the elements of Ext by actual module maps, rather than by using complexes.

1.B.4. *Some other examples.*

- The derived category $\mathsf{D}_{\mathrm{qc}}(X)$ of complexes of $\mathcal{O}_X$-modules with quasi-coherent cohomology for a nice enough scheme X.
- The (equivariant) stable homotopy category of spectra (representing objects for (equivariant) cohomology theories for topological spaces).
- The Kasparov category of separable G-C^*-algebras.

2. Tensor-triangulated categories

Recall that a symmetric monoidal structure on a category $\mathcal{C}$ is the data of a bifunctor

$$- \otimes - \colon \mathcal{C} \times \mathcal{C} \to \mathcal{C}$$

called the *tensor product* (or *monoidal product*) together with an unit object $\mathbb{1} \in \mathcal{C}$ satisfying:

- unitality: $\mathbb{1} \otimes X \simeq X$ for all $X \in \mathcal{C}$;
- associativity: $(X \otimes Y) \otimes Z \simeq X \otimes (Y \otimes Z)$ for all $X, Y, Z \in \mathcal{C}$;
- symmetry: $X \otimes Y \simeq Y \otimes X$ for all $X, Y \in \mathcal{C}$;

all satisfying various coherences that we won't make precise here. Such a monoidal structure is moreover called *closed*, if there is a bifunctor $\underline{\mathrm{Hom}}(-,-) \colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{C}$ called the *internal hom* such that $- \otimes X$ is left adjoint to $\underline{\mathrm{Hom}}(X, -)$. Again, this is all subject to various coherences that we will not make explicit. Note that $\underline{\mathrm{Hom}}(\mathbb{1}, X) \simeq X$ for all $X \in \mathcal{C}$.

In order to blend this data with that of a triangulated category we need one more definition:

Definition 2.1. Let T and U be triangulated categories. A *triangulated functor* is an additive functor $F \colon \mathsf{T} \to \mathsf{U}$ together with a natural isomorphism $\varphi \colon F\Sigma \xrightarrow{\sim} \Sigma F$ such that for any triangle

$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$ in T, the candidate triangle

$$FX \xrightarrow{F(f)} FY \xrightarrow{F(g)} FZ \xrightarrow{\varphi_X \circ F(h)} \Sigma FX$$

is a triangle in U.

With this, we can make the key definition:

Definition 2.2. A *tensor-triangulated category* T (tt-category for short) is a triangulated category which also has a closed symmetric monoidal structure which is compatible in the following sense: the tensor product is a triangulated functor in both variables, and the internal hom is a triangulated functor in both variables.

Remark 2.3. In fact, we only require that $\underline{\mathrm{Hom}}(-, X)$ is triangulated up to sign. We ignore this distinction, since for most purposes the signs are irrelevant.

Definition 2.4. A functor $F\colon \mathsf{T} \to \mathsf{U}$ between tt-categories is a *tt-functor* if it is triangulated and strong monoidal. (Recall that strong monoidality means that there are natural isomorphisms $F(X \otimes Y) \simeq FX \otimes FY$ and $F(\mathbb{1}_{\mathsf{T}}) \simeq \mathbb{1}_{\mathsf{U}}$ satisfying various coherences.)

2.A. **Examples.** We record two key examples.

2.A.1. *The derived category of a commutative ring.* If R is a commutative ring then $\mathsf{D}(R)$ is a tensor-triangulated category. The tensor product is given by the derived tensor product $- \otimes_R^{\mathsf{L}} -$ and the internal hom is given by the derived hom $\mathsf{RHom}_R(-,-)$. These are defined as follows. Note that for an arbitrary $M \in \mathsf{D}(R)$, the functors $M \otimes_R -$ and $\mathrm{Hom}_R(M,-)$ do not preserve quasi-isomorphisms, so cannot induce a functor on the derived category. This is the case even if M is a complex of projective R-modules:

Example 2.5. Consider $R = \mathbb{Z}/4$ and the complex

$$M = \cdots \to \mathbb{Z}/4 \xrightarrow{\cdot 2} \mathbb{Z}/4 \xrightarrow{\cdot 2} \mathbb{Z}/4 \to \cdots .$$

The quasi-isomorphism $M \xrightarrow{\sim} 0$ is not preserved by $\mathrm{Hom}_{\mathbb{Z}/4}(M,-)$.

Instead, we use *dg-projective resolutions*: a dg-projective resolution for $M \in \mathsf{D}(R)$ consists of a quasi-isomorphism $P \xrightarrow{\sim} M$ where P has the property that $\mathrm{Hom}_R(P,-)$ preserves surjective quasi-isomorphisms. The derived tensor product is then defined by $M \otimes_R^{\mathsf{L}} - = P \otimes_R -$ where P is a dg-projective resolution of M. Likewise $\mathsf{RHom}_R(M,-) = \mathrm{Hom}_R(P,-)$. There is an analogous notion of dg-injective resolution used to define $\mathsf{RHom}_R(-,M)$.

2.A.2. *The stable module category.* Let G be a finite group and k be a field of characteristic dividing the order of G. Then the stable module category $\mathsf{StMod}(kG)$ is a tensor-triangulated category. The tensor product is given by $- \otimes_k -$ with the diagonal G-action, with monoidal unit k with the trivial action. To see that this is indeed well-defined on the stable category, let us first consider what happens on $\mathsf{Mod}(kG)$. Firstly note that $M \otimes_k N \simeq N \otimes_k M$; one easily checks that this is indeed kG-linear. One can make $\mathrm{Hom}_k(M,N)$ into a kG-module via $(g \cdot f)(m) = gf(g^{-1}m)$. In particular, one sees that $g \cdot f = f$ for all $g \in G$ if and only if $f(gm) = gf(m)$ for all g, m. Therefore,

$$\mathrm{Hom}_k(M,N)^G = \mathrm{Hom}_{kG}(M,N).$$

From this, one checks that one has the tensor-hom adjunction

$$\mathrm{Hom}_{kG}(L \otimes_k M, N) = \mathrm{Hom}_{kG}(L, \mathrm{Hom}_k(M,N)).$$

From this the following is easy:

Lemma 2.6. *For any kG-module M, if P is a projective kG-module, then $P \otimes_k M$ is again projective.*

Proof. We have
$$\mathrm{Hom}_{kG}(P \otimes_k M, -) \simeq \mathrm{Hom}_{kG}(P, \mathrm{Hom}_k(M, -))$$
by the tensor-hom adjunction. The latter is an exact functor, hence $P \otimes_k M$ is projective. □

Since the tensor product on Mod_{kG} preserves projectives, it descends to a tensor product on $\mathsf{StMod}(kG)$. In a similar way one sees that $\mathrm{Hom}_k(M, -)$ preserves injectives, and hence also passes to give a functor on $\mathsf{StMod}(kG)$. Therefore, $\mathsf{StMod}(kG)$ is indeed a tt-category as claimed.

2.B. **Smallness in tt-categories.** Let us first briefly return to the setting of a triangulated category T (i.e., not necessarily with a tensor product). We study certain types of 'small' objects in triangulated categories. It is often the case that the whole triangulated category is generated by small objects, and much of the theory of triangulated categories relies upon such assumptions.

Definition 2.7. Let T be a triangulated category which has coproducts. An object $X \in \mathsf{T}$ is said to be *compact* if the natural map
$$\bigoplus \mathrm{Hom}_{\mathsf{T}}(X, Y_i) \to \mathrm{Hom}_{\mathsf{T}}(X, \bigoplus Y_i)$$
is an equivalence for every set of objects $\{Y_i\}$. We write T^{c} for the full subcategory of T consisting of the compact objects.

Before we can give some examples of compact objects, and criteria for detecting them, we must introduce some terminology.

Definition 2.8. Let T be a triangulated category which has coproducts.

(1) A full subcategory $\mathcal{S}$ of T is *thick* if it is closed under retracts and is triangulated.
(2) A full subcategory $\mathcal{S}$ of T is *localizing* if it is thick, and closed under coproducts.

Given a set of objects $\mathcal{X}$ of T, we write $\mathrm{Thick}(\mathcal{X})$ (resp., $\mathrm{Loc}(\mathcal{X})$) for the smallest thick (resp., localizing) subcategory of T containing $\mathcal{X}$. Note that these are well defined since the intersection of thick/localizing subcategories is again thick/localizing.

Example 2.9. Let $X \in \mathsf{T}$. The full subcategory $\{Y \in \mathsf{T} \mid \mathrm{Hom}_{\mathsf{T}}(\Sigma^i X, Y) \simeq 0 \text{ for all } i \in \mathbb{Z}\}$ is thick. It is localizing if X is compact.

Definition 2.10. Let $\mathcal{X}$ be a set of objects of T. We say that $\mathcal{X}$ *generates* T if $\mathrm{Loc}(\mathcal{X}) = \mathsf{T}$. If each element of $\mathcal{X}$ is compact, then we say that $\mathcal{X}$ *compactly generates* T.

Let us give an alternative characterisation of compact generation:

Proposition 2.11. *Let T be a triangulated category with coproducts, and $\mathcal{S}$ be a set of compact objects of T. Then the following are equivalent:*

(1) $\mathcal{S}$ generates T;
(2) if $\mathrm{Hom}_{\mathsf{T}}(\Sigma^i S, X) = 0$ for all $S \in \mathcal{S}$ and $i \in \mathbb{Z}$, then $X \simeq 0$.

Proof. Left as Exercise A.1. □

If T is compactly generated, then we can characterise the compact objects in terms of building operations.

Proposition 2.12. *Let* T *be a triangulated category which is compactly generated by a set* G*. Then* $\mathsf{T}^{\mathrm{c}} = \mathrm{Thick}(\mathsf{G})$*.*

Proof. The implication that $X \in \mathsf{T}^{\mathrm{c}}$ implies that $X \in \mathrm{Thick}(\mathsf{G})$ requires some work so we omit it. For the converse, consider the set

$$\{Y \in \mathsf{T} \mid \bigoplus \mathrm{Hom}_{\mathsf{T}}(Y, Z_i) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{T}}(Y, \bigoplus Z_i) \text{ for all sets } \{Z_i\}\}.$$

This is a thick subcategory of T, and contains G. Hence it contains $\mathrm{Thick}(\mathsf{G})$ since this is the *smallest* thick subcategory of T containing G. □

Example 2.13. We have $\mathsf{D}(R)^{\mathrm{c}} = \mathsf{Perf}(R)$ where the right hand side denotes the full subcategory of *perfect complexes*, that is, those complexes which are quasi-isomorphic to a bounded complex of finitely generated projectives. We leave the proof of this as Exercise A.2.

Example 2.14. We have $\mathsf{StMod}(kG)^{\mathrm{c}} = \mathsf{stmod}(kG)$ where the right hand side denotes the full subcategory of finitely generated modules. If M is compact, then writing M as a direct limit $M = \varinjlim M_i$ of finitely generated modules, we have $\mathrm{Hom}(M, M) = \varinjlim \mathrm{Hom}(M, M_i)$. In particular, the identity on M factors through M_i so that M_i is a retract of a finitely generated and hence is finitely generated. The other inclusion is left as Exercise A.3.

If T is a tt-category, one might hope that T^{c} inherits these properties. Unfortunately, this need not always be the case: the tensor product of compacts need not be compact in general, and the unit need not be compact in general.

Example 2.15. Let $\mathsf{T} = \prod_i \mathsf{T}_i$ be the countable product of tt-categories T_i with tt-structure given pointwise. Then a tuple $(X_i) \in \mathsf{T}$ is compact if and only if for almost all i we have $X_i = 0$ and $X_j \in \mathsf{T}_j^c$ for all j. In particular, since $\mathbb{1}_{\mathsf{T}} = (\mathbb{1}_{\mathsf{T}_i})$ the unit is not compact.

There is an alternative subcategory of 'small' object which behaves better with the tensor structure: Recall that the counit of the tensor-hom adjunction is the evaluation map $\mathrm{ev}\colon X \otimes \underline{\mathrm{Hom}}(X, Y) \to Y$ and the unit is the coevaluation map $\mathrm{coev}\colon Y \to \underline{\mathrm{Hom}}(X, X \otimes Y)$.

Definition 2.16. Suppose that T is a tensor-triangulated category with coproducts.

(1) An object $X \in \mathsf{T}$ is said to be *rigid* if the natural map

$$\nu_{X,Y}\colon \underline{\mathrm{Hom}}(X, \mathbb{1}) \otimes Y \to \underline{\mathrm{Hom}}(X, Y)$$

is an equivalence for all $Y \in \mathsf{T}$. To spell it out, the natural map is the composite

$$\underline{\mathrm{Hom}}(X, \mathbb{1}) \otimes Y \xrightarrow{\mathrm{coev}} \underline{\mathrm{Hom}}(X, X \otimes \underline{\mathrm{Hom}}(X, \mathbb{1}) \otimes Y) \xrightarrow{\underline{\mathrm{Hom}}(X, \mathrm{ev})} \underline{\mathrm{Hom}}(X, Y).$$

(2) An object $X \in \mathsf{T}$ is said to be *internally compact* if the natural map

$$\bigoplus \underline{\mathrm{Hom}}(X, Y_i) \to \underline{\mathrm{Hom}}(X, \bigoplus Y_i)$$

is an equivalence for every set $\{Y_i\}$.

Example 2.17. For any tt-category T, the tensor unit $\mathbb{1}$ is both rigid and internally compact.

In order to study the properties of rigid objects it is helpful to consider an alternative definition (which we will show is in fact equivalent). We write $D\colon \mathsf{T}^{\mathrm{op}} \to \mathsf{T}$ for the functor $D = \underline{\mathrm{Hom}}(-, \mathbb{1})$. This is called the *functional dual* (or sometimes Spanier-Whitehead dual).

Definition 2.18. Let T be a tensor-triangulated category, and $X \in \mathsf{T}$. The object X is *dualizable* if and only if there exists a map $\eta\colon \mathbb{1} \to X \otimes DX$ such that the diagram

$$\begin{tikzcd} \mathbb{1} \arrow[r, "\eta"] \arrow[dr, "\mathrm{coev}"'] & X \otimes DX \arrow[d, "\nu_{X,X}"] \\ & \underline{\mathrm{Hom}}(X, X) \end{tikzcd}$$

commutes.

Firstly, note that there is a natural map $\rho\colon X \to D^2X$ given by the composite

$$X \xrightarrow{\mathrm{coev}} \underline{\mathrm{Hom}}(DX, X \otimes DX) \xrightarrow{\underline{\mathrm{Hom}}(DX,\mathrm{ev})} \underline{\mathrm{Hom}}(DX, \mathbb{1})$$

for all $X \in \mathsf{T}$. Many of the proofs of the following statements follow by tedious diagram chases, so we provide only an outline of the proofs.

Lemma 2.19. *Let T be a tensor-triangulated category and $X \in \mathsf{T}$. If X is dualizable then DX is dualizable.*

Proof. We define a map $\mathbb{1} \to DX \otimes D^2X$ via the composite

$$\mathbb{1} \xrightarrow{\eta} X \otimes DX \xrightarrow{\rho \otimes DX} D^2X \otimes DX.$$

It remains to check that the required diagram commutes. That is, we want to check that the outer square in the diagram

$$\begin{tikzcd} \mathbb{1} \arrow[rr, "\eta"] \arrow[dr, "\mathrm{coev}"] \arrow[dd, "\mathrm{coev}"'] & & X \otimes DX \arrow[dl, "\nu_{X,X}"'] \arrow[dd, "\rho \otimes DX"] \\ & \underline{\mathrm{Hom}}(X, X) \arrow[dl, "\alpha"'] & \\ \underline{\mathrm{Hom}}(DX, DX) & & D^2X \otimes DX \arrow[ll, "\nu_{DX,DX}"] \end{tikzcd}$$

commutes, where α is defined to be the composite

$$\underline{\mathrm{Hom}}(X, X) \xrightarrow{\underline{\mathrm{Hom}}(X,\rho)} \underline{\mathrm{Hom}}(X, D^2X) \simeq \underline{\mathrm{Hom}}(X \otimes DX, \mathbb{1}) \simeq \underline{\mathrm{Hom}}(DX, DX).$$

The top triangle commutes since X is dualizable, so it suffices to prove that the other subdiagrams commute. This can be checked by unravelling the definitions of α and ρ; we leave this as a (tedious) exercise for the reader. □

Lemma 2.20. *Let T be a tensor-triangulated category, and suppose that $X \in \mathsf{T}$ is dualizable. Then the natural map $\rho\colon X \to D^2X$ is an equivalence.*

Proof. We claim that the composite

$$D^2X \simeq \mathbb{1} \otimes D^2X \xrightarrow{\eta \otimes D^2X} X \otimes DX \otimes D^2X \xrightarrow{X \otimes \mathrm{ev}} X$$

is an inverse to ρ. This can be checked by diagram chasing. □

Proposition 2.21. *Let T be a tensor-triangulated category, and $X \in \mathsf{T}$. Then X is rigid if and only if it is dualizable.*

Proof. The forward direction is clear since we may take $\eta = \nu_{X,X}^{-1} \circ \mathrm{coev}$. For the reverse direction, we claim that the composite

$$\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(X, Y) \otimes \mathbb{1} \xrightarrow{\underline{\mathrm{Hom}}(X,Y) \otimes \eta} \underline{\mathrm{Hom}}(X, Y) \otimes X \otimes DX \xrightarrow{\mathrm{ev} \otimes DX} Y \otimes DX$$

is an inverse to $\nu_{X,Y}$. Again, one may verify this by diagram chasing. □

Remark 2.22. By Lemma 2.20, one deduces that if X is a rigid object, then we have adjunctions $X \otimes - \dashv DX \otimes - \dashv X \otimes -$.

Lemma 2.23. *Let $F\colon \mathsf{T} \to \mathsf{U}$ be a tt-functor between tt-categories. Then F preserves rigid (=dualizable) objects. Moreover, if X is rigid, then $F(DX) \simeq D(FX)$.*

Proof. For any $X \in \mathsf{T}$, we can construct a map $\varphi\colon F(DX) \to D(FX)$ as the adjunct to

$$FX \otimes F(DX) \simeq F(X \otimes DX) \xrightarrow{F(\mathrm{ev})} F(\mathbb{1}_{\mathsf{T}}) \simeq \mathbb{1}_{\mathsf{U}}.$$

If X is dualisable, then we have a map $\eta\colon \mathbb{1}_{\mathsf{T}} \to X \otimes DX$ and hence we have a map $\eta'\colon \mathbb{1}_{\mathsf{U}} \to FX \otimes D(FX)$ defined as the composite

$$\mathbb{1}_{\mathsf{U}} \simeq F(\mathbb{1}_{\mathsf{T}}) \xrightarrow{F(\eta)} F(X \otimes DX) \simeq FX \otimes F(DX) \xrightarrow{FX \otimes \varphi} FX \otimes D(FX).$$

Using this one can verify that FX is dualisable.

To see that the map $\varphi\colon F(DX) \to D(FX)$ constructed above is an isomorphism, one can construct an inverse as follows. Since FX is dualisable, $D(FX)$ is also dualisable (Lemma 2.19) so by adjunction maps $D(FX) \to F(DX)$ correspond to maps $\mathbb{1}_{\mathsf{U}} \to FX \otimes F(DX)$. We can construct such a map as the composite

$$\mathbb{1}_{\mathsf{U}} \simeq F(\mathbb{1}_{\mathsf{T}}) \xrightarrow{F(\eta)} F(X \otimes DX) \simeq FX \otimes F(DX).$$

One can check directly that this gives an inverse to φ. □

Is it natural to ask how the different notions of 'smallness' in Definition 2.7 and Definition 2.16 are related. The following two results answer this. We also note the following relation, which does not rely on a set of rigid generators.

Proposition 2.24. *Let T be a tensor-triangulated category. If $\mathbb{1}$ is compact, then any rigid object is compact.*

Proof. Let X be rigid. Then $\mathrm{Hom}(X, -) \simeq \mathrm{Hom}(\mathbb{1}, DX \otimes -)$. Since $\mathbb{1}$ is compact this commutes with coproducts showing that X is compact. □

Proposition 2.25. *Let T be a tensor-triangulated category, and suppose that T has a set of rigid generators G.*

(1) For $X \in \mathsf{T}$ we have

$$X \in \mathrm{Thick}(\mathsf{G}) \implies X \text{ is rigid} \iff X \text{ is internally compact.}$$

(2) If the elements of G are compact, then for $X \in \mathsf{T}$ we have

$$X \text{ is compact} \iff X \in \mathrm{Thick}(\mathsf{G}) \implies X \text{ is rigid} \iff X \text{ is internally compact.}$$

(3) If the elements of G are compact, and the unit $\mathbb{1}$ of T is compact, then for $X \in \mathsf{T}$ we have

$$X \text{ is compact} \iff X \in \mathrm{Thick}(\mathsf{G}) \iff X \text{ is rigid} \iff X \text{ is internally compact.}$$

Proof. For (1), firstly note that the set of rigid objects in T is thick. By assumption it contains G, so we have $\mathrm{Thick}(\mathsf{G}) \subseteq \{\text{rigid objects}\}$, which proves the first implication. That rigid objects are internally compact is an immediate consequence of the definitions. For the remaining implication, suppose X is internally compact, and consider the set

$$\mathcal{L} = \{Y \in \mathsf{T} \mid DX \otimes Y \xrightarrow{\sim} \underline{\mathrm{Hom}}(X, Y)\}.$$

The set $\mathcal{L}$ is localizing as X is internally compact, and contains G by Exercise A.4. Therefore $\mathcal{L} = \mathsf{T}$, and hence X is rigid.

For (2), it suffices to prove that X is compact if and only if $X \in \mathrm{Thick}(\mathsf{G})$, which was the content of Proposition 2.12. (3) follows from Proposition 2.24. □

Definition 2.26. Let T be a compactly generated tt-category. We say that T is *rigidly-compactly generated* if the compact and rigid objects in T coincide. Note that by Proposition 2.25(3) this is equivalent to T having a set of rigid and compact generators and the tensor unit being compact.

Example 2.27. Let T be a tt-category which is compactly generated by its tensor unit. Then T is rigidly-compactly generated. For example, $\mathsf{D}(R)$ is rigidly-compactly generated.

Example 2.28. The stable module category $\mathsf{StMod}(kG)$ is rigidly-compactly generated. Since the compacts are the finitely generated modules it is clear that it is compactly generated. It is easy to see that the compacts coincide with the rigid objects (since if M is finitely generated over kG then it has finite rank over k).

Example 2.29. Let us revisit Example 2.15: we consider $\mathsf{T} = \prod_i \mathsf{T}_i$ where each T_i is rigidly-compactly generated by G_i. The set $\mathsf{G} = \{\delta_i(G_i) \mid G_i \in \mathsf{G}_i\}$ of "characteristic objects" given by $(\delta_i X)_i = X_i$ and $(\delta_i X)_j = 0$ for $i \neq j$ is a set of rigid generators for T. Moreover these are compact. So we are in case (2) of Proposition 2.25. However, we already saw that the unit is not compact (but is rigid). So the inclusion $\mathsf{T}^{\mathrm{c}} \subset \{\text{rigids}\}$ is proper.

Finally, it is convenient to introduce terminology for the appropriate analogues of thick and localizing subcategories in the tensor-triangular setting.

Definition 2.30. Let T be a tensor-triangulated category. A thick/localizing subcategory $\mathcal{S}$ is a *$\otimes$-ideal* if for all $X \in \mathcal{S}$ and $Y \in \mathsf{T}$ we have $X \otimes Y \in \mathcal{S}$. We write $\mathrm{Thick}^{\otimes}(-)$ and $\mathrm{Loc}^{\otimes}(-)$ for the smallest thick and localizing $\otimes$-ideal generated by a set of objects.

Proposition 2.31. *Let T be a tensor-triangulated category and $X \in \mathsf{T}$ be rigid.*

(1) If Y is compact, then $X \otimes Y$ is compact.
(2) If Y is internally compact, then $X \otimes Y$ is internally compact.
(3) If Y is rigid, then $X \otimes Y$ is rigid.

Proof. For (1), we isomorphisms

$$\mathrm{Hom}(X \otimes Y, \bigoplus Z_i) = \mathrm{Hom}(X, DY \otimes \bigoplus Z_i) = \bigoplus \mathrm{Hom}(X, DY \otimes Z_i) = \bigoplus \mathrm{Hom}(X \otimes Y, Z_i)$$

showing that $X \otimes Y$ is compact. Parts (2) and (3) are similar and are left as Exercise A.5. □

The following is then immediate:

Corollary 2.32. *For any tensor-triangulated category T, the full subcategory of rigid objects forms a tensor-triangulated category.* □

3. Geometric functors

In this section we introduce the notion of a *geometric functor*. In order to demonstrate their powerful formal properties, we recall the adjoint functor theorems for triangulated categories.

Lemma 3.1. *Let $F\colon \mathsf{T} \to \mathsf{U}$ be a triangulated functor. If F has a adjoint (on either side), then this adjoint is also a triangulated functor.*

Proof. Left as Exercise A.9. □

Theorem 3.2 (Adjoint functor theorems)**.** *Let T and U be triangulated categories. Let $F\colon \mathsf{T} \to \mathsf{U}$ be a triangulated functor, and suppose that T is compactly generated.*

(1) F is coproduct preserving if and only if F has a triangulated right adjoint.
(2) F is product preserving if and only if F has a triangulated left adjoint.

Proof. This is an immediate consequence of Brown representability and Lemma 3.1. □

Proposition 3.3. *Let $F : \mathsf{T} \rightleftarrows \mathsf{U} : G$ be an adjunction between triangulated categories.*

(1) If G preserves coproducts, then F preserves compact objects.
(2) Suppose that T is compactly generated and that F preserves compacts. Then G preserves coproducts.

Proof. For (1) we let $C \in \mathsf{T}^{\mathrm{c}}$. For any set of objects $\{X_i\}$ in U, we have isomorphisms

$$\oplus\mathrm{Hom}(FC, X_i) \simeq \oplus\mathrm{Hom}(C, GX_i) \simeq \mathrm{Hom}(C, \oplus GX_i) \simeq \mathrm{Hom}(C, G(\oplus X_i)) \simeq \mathrm{Hom}(FC, \oplus X_i)$$

using in turn, the adjunction, compactness of C, the assumption that G preserves coproducts, and finally the adjunction again. For (2), by compact generation of T it suffices to prove that the canonical map $\oplus GX_i \to G(\oplus X_i)$ induces isomorphisms

$$\mathrm{Hom}(C, \oplus GX_i) \xrightarrow{\simeq} \mathrm{Hom}(C, G(\oplus X_i))$$

for all $C \in \mathsf{T}^{\mathrm{c}}$. This is then a straightforward check using the adjunction and the fact that F preserves compacts. □

We now have the necessary triangulated background to make the key definition and prove some key features of it. We will then record several examples afterwards.

Definition 3.4. Let T and U be rigidly-compactly generated tt-categories. A *geometric functor* $f^*\colon \mathsf{T} \to \mathsf{U}$ is a strong symmetric monoidal, triangulated functor which preserves coproducts.

Proposition 3.5. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. Then f^* fits into an adjoint triple*

$$\begin{tikzcd} \mathsf{T} \arrow[r, bend left=50, "f^*"] \arrow[r, bend right=50, "f^{(1)}"'] & \mathsf{U} \arrow[l, "f_*"'] \end{tikzcd}$$

where left adjoints are denoted on top.

Proof. Since f^* is coproduct preserving, it has a right adjoint f_* by Theorem 3.2. Since f^* is a strong monoidal functor it preserves rigid objects (Lemma 2.23) and hence preserves compact objects since these coincide by assumption. Therefore, f_* preserves coproducts by Proposition 3.3 and hence itself has a right adjoint $f^{(1)}$ again by Theorem 3.2. □

Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. There is a natural map $\pi\colon X \otimes f_*Y \to f_*(f^*X \otimes Y)$ for all $X \in \mathsf{T}$ and $Y \in \mathsf{U}$ which is constructed as the adjunct to the natural map

$$f^*(X \otimes f_*Y) \simeq f^*X \otimes f^*f_*Y \xrightarrow{f^*X \otimes \varepsilon} f^*X \otimes Y$$

where ε is the counit of the (f^*, f_*) adjunction.

Proposition 3.6 (The projection formula)**.** *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. The natural map*

$$\pi\colon X \otimes f_*Y \xrightarrow{\simeq} f_*(f^*X \otimes Y)$$

is an isomorphism for all $X \in \mathsf{T}$ and $Y \in \mathsf{U}$.

Proof. Let $Z \in \mathsf{T}$ and $Y \in \mathsf{U}$ be arbitrary. We first prove that π is an isomorphism if X is rigid. In this case, we have isomorphisms

$$\begin{aligned}
\mathrm{Hom}(Z, X \otimes f_*Y) &\simeq \mathrm{Hom}(Z \otimes DX, f_*Y) && \text{since } X \text{ is rigid} \\
&\simeq \mathrm{Hom}(f^*Z \otimes f^*(DX), Y) && \text{by adjunction and monoidality of } f^* \\
&\simeq \mathrm{Hom}(f^*Z \otimes D(f^*X), Y) && \text{by Lemma 2.23} \\
&\simeq \mathrm{Hom}(Z, f_*(f^*X \otimes Y)) && \text{by rigidity of } D(f^*X) \text{ and adjunction.}
\end{aligned}$$

A diagram chase verifies that this isomorphism is induced by π (we omit tedious manipulations of commutative diagrams throughout in this course). By the Yoneda lemma we therefore deduce that π is an isomorphism when X is rigid. Now the collection of $X \in \mathsf{T}$ for which π is an isomorphism (for a fixed Y) is a localising subcategory (since both f^* and f_* preserve coproducts and are triangulated). This localising subcategory contains all the rigid=compact objects, and hence by compact generation, we deduce that π is an isomorphism for all X. □

A slick trick allows one to generate new isomorphisms from these by using uniqueness of adjunctions. Recall that if $F_1 \dashv G_1$ and $F_2 \dashv G_2$, then $F_1F_2 \dashv G_2G_1$ (notice the order reversal).

Proposition 3.7. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. There are natural isomorphisms:*

$$\underline{\mathrm{Hom}}(X, f_*Y) \simeq f_*\underline{\mathrm{Hom}}(f^*X, Y) \tag{3.8}$$

$$\underline{\mathrm{Hom}}(f_*X, Y) \simeq f_*\underline{\mathrm{Hom}}(X, f^{(1)}Y) \tag{3.9}$$

$$f^{(1)}\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(f^*X, f^{(1)}Y). \tag{3.10}$$

Proof. We first fix X in the projection formula (Proposition 3.6). We have adjunctions:

$$X \otimes f_*(-) = (X \otimes -) \circ f_* \dashv f^{(1)} \circ \underline{\mathrm{Hom}}(X, -)$$

and

$$f_*(f^*X \otimes -) \dashv \underline{\mathrm{Hom}}(f^*X, -) \circ f^{(1)}.$$

By uniqueness of adjoints, we therefore deduce that $f^{(1)}\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(f^*X, f^{(1)}Y)$, giving (3.10). Fixing Y in the projection formula instead yields (3.9). Finally, for (3.8), we have an isomorphism $f^*X \otimes f^*(-) \simeq f^*(X \otimes -)$. Taking right adjoints of this gives the claim. □

Let us return to the examples and see what the above features of geometric functors correspond to in this setting.

Example 3.11. Let $f \colon R \to S$ be a map of commutative rings. Then the extension of scalars functors $f^* = S \otimes^{\mathsf{L}}_R - \colon \mathsf{D}(R) \to \mathsf{D}(S)$ is a geometric functor. Its right adjoint f_* is the restriction of scalars functor, and the double right adjoint is the coextension of scalars functor $f^{(1)} = \mathsf{RHom}_R(S, -)$. In this case, the projection formula is not exciting: it says that for $X \in \mathsf{D}(R)$ and $Y \in \mathsf{D}(S)$ there is an isomorphism $X \otimes^{\mathsf{L}}_R Y \simeq (S \otimes^{\mathsf{L}}_R X) \otimes^{\mathsf{L}}_S Y$. The other isomorphisms are similarly "obvious" in this setting. The main selling point of the framework is that these are general isomorphisms which hold in all settings, and moreover admit uniform proofs based purely on formal manipulations.

Example 3.12. Let G be a group and H be a subgroup. There is a restriction functor $f^* = \mathrm{res}^G_H \colon \mathsf{StMod}(kG) \to \mathsf{StMod}(kH)$ given by restricting the G-action to a H-action, and this is a geometric functor. This has a right adjoint given by coinduction $f_* = \mathrm{Hom}_{kH}(kG, -)$ which has a further right adjoint given by restriction again. In this example, we see some extra features happening: f^* also has a left adjoint (the induction $kG \otimes_{kH} -$) and this left adjoint is isomorphic to the coinduction. Therefore we obtain an infinite tower of adjunctions. This is an instance of the *Wirthmüller isomorphism* which we will see later on.

Remark 3.13. We see that the extra features visible in Example 3.12 are not uniform for all geometric functors. Indeed, they do not occur in the setting of Example 3.11 in general. However, one can completely characterise the existence of extra adjoints and when these are isomorphic in general tt-terms. This is the point of much of the rest of this course.

4. Grothendieck–Neeman duality

Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. In this section, we explain when f^* and its double right adjoint $f^{(1)}$ agree up to a twist. Such an isomorphism is termed *Grothendieck duality*. Let us first give the background to make this precise.

Definition 4.1. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. We set $\omega_f \coloneqq f^{(1)}(\mathbb{1}_{\mathsf{T}})$ and call this the *relative dualising object*.

Remark 4.2. For any $X, Y \in \mathsf{T}$ there is a natural map

$$f^{(1)}Y \otimes f^*X \to f^{(1)}(Y \otimes X)$$

constructed as the adjoint to $f_*(f^{(1)}Y \otimes f^*X) \simeq f_*f^{(1)}Y \otimes X \xrightarrow{\varepsilon \otimes X} Y \otimes X$ where the isomorphism is the projection formula (Proposition 3.6) and ε is the counit map. Taking $Y = \mathbb{1}_{\mathsf{T}}$ we obtain a natural map $\omega_f \otimes f^*X \to f^{(1)}X$. When this map is an isomorphism, we say that *Grothendieck duality* holds for X.

We first show that all rigid objects have Grothendieck duality:

Lemma 4.3. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. If $X \in \mathsf{T}$ is rigid, then the natural map*

$$f^{(1)}Y \otimes f^*X \to f^{(1)}(Y \otimes X)$$

is an isomorphism for all $Y \in \mathsf{T}$.

Proof. Let $Z \in \mathsf{T}$ be arbitrary. We have natural isomorphisms:

$$\begin{aligned}
\mathrm{Hom}(Z, f^{(1)}Y \otimes f^*X) &\simeq \mathrm{Hom}(Z \otimes f^*(DX), f^{(1)}Y) && \text{by Lemma 2.23}\\
&\simeq \mathrm{Hom}(f_*Z \otimes DX, Y) && \text{by adjunction and Proposition 3.6}\\
&\simeq \mathrm{Hom}(f_*Z, X \otimes Y) && \text{as } X \text{ is rigid}\\
&\simeq \mathrm{Hom}(Z, f^{(1)}(X \otimes Y)) && \text{by adjunction.}
\end{aligned}$$

A diagram chase verifies this is induced by the natural map $f^{(1)}Y \otimes f^*X \to f^{(1)}(Y \otimes X)$ and hence by the Yoneda lemma we are done. □

The goal is to generalise this in two directions:

(1) We identify conditions when every object of T has Grothendieck duality.
(2) Considering some examples in commutative algebra, one sees that the conditions we will identify are rather strong in some cases, so we will give an alternative condition which yields Grothendieck duality on an enlargement of the subcategory of rigid objects.

In this section we deal with (1). The next few sections treat (2).

Remark 4.4. Notice that if every object has Grothendieck duality, then $f^{(1)}$ preserves coproducts. Conversely, if $f^{(1)}$ preserves coproducts then every object has Grothendieck duality by a localising subcategory argument. We now identify further equivalent conditions in terms of the more familiar functors f^* and f_* (where more familiar means in the sense of explicit examples).

We first prove a general result about triangulated categories which provides a way to determine properties of a compact-preserving functor by restriction to compacts.

Lemma 4.5. *Let $F : \mathsf{T} \rightleftarrows \mathsf{U} : G$ be an adjunction of triangulated functors between triangulated categories, where T is compactly generated. Suppose that F preserves compacts. If the restriction $F|\colon \mathsf{T}^c \to \mathsf{U}^c$ has a right adjoint G_0, then G preserves compacts. Moreover, the restriction of G to compacts is G_0.*

Proof. We have isomorphisms

$$\mathrm{Hom}_{\mathsf{T}^c}(X, G_0(Y)) = \mathrm{Hom}_{\mathsf{U}^c}(F|(X), Y) = \mathrm{Hom}_{\mathsf{U}}(FX, Y) = \mathrm{Hom}_{\mathsf{T}}(X, GY)$$

for all $X \in \mathsf{T}^c$ and $Y \in \mathsf{U}^c$. Therefore taking $X = G_0(Y)$, we obtain a natural map $g\colon G_0(Y) \to GY$ corresponding to the identity on $G_0(Y)$. One checks that the above isomorphism of Hom's is induced by $\mathrm{Hom}_{\mathsf{T}}(-, g)$ and hence by compact generation of T we deduce that g is an isomorphism which gives the claims. □

Lemma 4.6. *Let $F : \mathsf{T} \rightleftarrows \mathsf{U} : G$ be an adjunction of triangulated functors between triangulated categories, where both T and U are compactly generated. Suppose that F preserves compacts. If the restriction $F|\colon \mathsf{T}^c \to \mathsf{U}^c$ has a left adjoint E_0, then F preserves products.*

Proof. Let $X \in \mathsf{U}^c$ and $Y \in \mathsf{T}$. There is a natural map $\alpha_{X,Y}\colon \mathrm{Hom}(E_0(X), Y) \to \mathrm{Hom}(X, FY)$ defined by the composite

$$\mathrm{Hom}(E_0(X), Y) \xrightarrow{F} \mathrm{Hom}(FE_0(X), FY) \xrightarrow{\mathrm{Hom}(\eta, FY)} \mathrm{Hom}(X, FY)$$

which is an isomorphism if Y is compact. Notice that for a fixed $X \in \mathsf{U}^c$, the collection of $Y \in \mathsf{T}$ for which $\alpha_{X,Y}$ is an isomorphism is a localising subcategory by the 5-lemma and the fact that F preserves coproducts. Therefore by compact generation of T, we have that $\alpha_{X,Y}$ is an isomorphism for all $X \in \mathsf{U}^c$ and $Y \in \mathsf{T}$.

Let $X \in \mathsf{U}^c$ and $\{Y_i\} \subset \mathsf{T}$. We then have isomorphisms

$$\begin{aligned}
\mathrm{Hom}(X, F(\textstyle\prod Y_i)) &= \mathrm{Hom}(E_0(X), \textstyle\prod Y_i) \\
&= \textstyle\prod \mathrm{Hom}(E_0(X), Y_i) \\
&= \textstyle\prod \mathrm{Hom}(X, FY_i) \\
&= \mathrm{Hom}(X, \textstyle\prod (FY_i))
\end{aligned}$$

using α in the first and third isomorphism. One can check that this is induced by the canonical comparison map $F(\prod Y_i) \to \prod FY_i$ and hence by compact generation of U we deduce that F preserves products. □

Lemma 4.7. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. If f^* has a left adjoint $f_{(1)}$ then the following hold:*

(1) f_* preserves compact objects;
(2) if $X \in \mathsf{U}^c$, then there is a natural isomorphism $D(f_{(1)}(DX)) \simeq f_*X$.

Proof. Since f^* preserves coproducts, its left adjoint $f_{(1)}$ preserves compacts by Proposition 3.3. Now by Lemma 2.23 we have a commutative diagram

$$\begin{tikzcd}
(\mathsf{T}^c)^{\mathrm{op}} \arrow[r, "D"] \arrow[d, "(f^*)^{\mathrm{op}}"'] & \mathsf{T}^c \arrow[d, "f^*"] \\
(\mathsf{U}^c)^{\mathrm{op}} \arrow[r, "D"'] & \mathsf{U}^c.
\end{tikzcd}$$

Now notice that the *right* adjoint of $(f^*)^{\mathrm{op}}$ is $(f_{(1)})^{\mathrm{op}}$. As such (since D is an equivalence of categories) we see that $D(f_{(1)}^{\mathrm{op}}(D(-)))$ is right adjoint to f^*. Therefore by Lemma 4.5 we see that f_* preserves compacts and the restriction of f_* to compacts is $D(f_{(1)}(D(-)))$. □

Theorem 4.8. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. The following conditions are equivalent:*

(1) f^* has a left adjoint $f_{(1)}$;
(2) f^* preserves products;
(3) $f^{(1)}$ has a right adjoint $f_{(-1)}$;
(4) $f^{(1)}$ preserves coproducts;
(5) f_* preserves compacts.

If any of the above equivalent conditions hold we say that f^* satisfies Grothendieck–Neeman duality*.*

Proof. We have that (1) ⇔ (2) and (3) ⇔ (4) by Theorem 3.2. We also know that (4) ⇔ (5) by Proposition 3.3. We have (1) ⇒ (5) by Lemma 4.7. Finally, (5) ⇒ (2) is left as Exercise A.10. □

Remark 4.9. If f^* has Grothendieck–Neeman duality, then we have an adjoint quintuple

$$f_{(1)} \dashv f^* \dashv f_* \dashv f^{(1)} \dashv f_{(-1)}.$$

Grothendieck–Neeman duality provides a plethora of natural isomorphisms relating all of the functors in the associated adjoint quintuple:

Corollary 4.10. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor which satisfies Grothendieck–Neeman duality. Then there are natural isomorphisms on all objects:*

$$f^*X \otimes f^{(1)}Y \simeq f^{(1)}(X \otimes Y) \tag{4.11}$$
$$\omega_f \otimes f^*X \simeq f^{(1)}X \tag{4.12}$$
$$\underline{\mathrm{Hom}}(X, f_{(-1)}Y) \simeq f_*\underline{\mathrm{Hom}}(f^{(1)}X, Y) \tag{4.13}$$
$$\underline{\mathrm{Hom}}(X, f_{(-1)}Y) \simeq f_{(-1)}\underline{\mathrm{Hom}}(f^*X, Y) \tag{4.14}$$
$$f_{(-1)} \simeq f_*\underline{\mathrm{Hom}}(\omega_f, -) \tag{4.15}$$
$$f_{(1)} \simeq f_*(\omega_f \otimes -) \tag{4.16}$$
$$f^* \simeq \underline{\mathrm{Hom}}(\omega_f, f^{(1)}(-)) \tag{4.17}$$
$$\mathbb{1}_{\mathsf{U}} \simeq \underline{\mathrm{Hom}}(\omega_f, \omega_f) \tag{4.18}$$
$$X \otimes f_{(1)}Y \simeq f_{(1)}(f^*X \otimes Y) \tag{4.19}$$
$$f^*\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(f^*X, f^*Y) \tag{4.20}$$
$$\underline{\mathrm{Hom}}(f_{(1)}X, Y) \simeq f_*\underline{\mathrm{Hom}}(X, f^*Y) \tag{4.21}$$

Remark 4.22. Before we give the proof, let us name the most important of the above isomorphisms. We already coined (4.12) Grothendieck duality. (4.16) is called the ur-Wirthmüller isomorphism, and (4.19) is called the left projection formula.

Proof. We prove these in order:

(4.11) Fix an arbitrary $Y \in \mathsf{T}$. Notice that the subcategory

$$\{X \in \mathsf{T} \mid f^{(1)}Y \otimes f^*X \xrightarrow{\sim} f^{(1)}(Y \otimes X) \text{ is an isomorphism}\}$$

is a localising subcategory since $f^{(1)}$ preserves coproducts by Grothendieck–Neeman duality (and f^* does since it is geometric). By Lemma 4.3 this subcategory contains all rigid=compact objects, and hence is the whole of T.

(4.12) This is the case $Y = \mathbb{1}_{\mathsf{T}}$ of (4.11).

(4.13) We fix Y in (4.11) and notice that we have adjunctions

$$f^*(-) \otimes f^{(1)}Y \dashv f_*\underline{\mathrm{Hom}}(f^{(1)}Y, -) \quad \text{and} \quad f^{(1)}(Y \otimes -) \dashv \underline{\mathrm{Hom}}(-, f_{(-1)}Y)$$

so we obtain the claim from (4.11) by taking right adjoints after fixing Y.

(4.14) This follows from (4.11) by taking right adjoints after fixing X.

(4.15) This is the case $X = \mathbb{1}_{\mathsf{T}}$ of (4.13).

(4.16) We first construct the natural map. By adjunction, maps $f_*(\omega_f \otimes X) \to f_{(1)}X$ are in bijection with maps $\omega_f \otimes X \to f^{(1)}f_{(1)}X$ and we have such a natural map by taking the composite

$$\omega_f \otimes X \xrightarrow{\omega_f \otimes \eta} \omega_f \otimes f^* f_{(1)}X \xrightarrow{\sim} f^{(1)}f_{(1)}X$$

where the isomorphism is Grothendieck duality (4.12). We note that the collection of X for which the natural map $f_*(\omega_f \otimes X) \to f_{(1)}X$ is an isomorphism is a localising subcategory (since all the functors involved preserve coproducts). Therefore it suffices to prove it is an isomorphism for X rigid. So let $X \in \mathsf{U}$ be rigid and $Z \in \mathsf{T}$ be arbitrary. We then have isomorphisms

$$\begin{aligned}
\mathrm{Hom}(Z, f_*(\omega_f \otimes X)) &\simeq \mathrm{Hom}(f^*Z \otimes DX, \omega_f) && \text{by adjunction and rigidity of } X \\
&\simeq \mathrm{Hom}(f_*(f^*Z \otimes DX), \mathbb{1}_{\mathsf{T}}) && \text{by adjunction as } \omega_f = f^{(1)}(\mathbb{1}_{\mathsf{T}}) \\
&\simeq \mathrm{Hom}(Z \otimes f_*(DX), \mathbb{1}_{\mathsf{T}}) && \text{by projection formula (3.6)} \\
&\simeq \mathrm{Hom}(Z \otimes D(f_{(1)}X), \mathbb{1}_{\mathsf{T}}) && \text{by Lemma 4.7} \\
&\simeq \mathrm{Hom}(Z, f_{(1)}X) && \text{as } f_{(1)}X \text{ is rigid.}
\end{aligned}$$

A (rather involved) diagram chase shows that this isomorphism is induced by the natural map $f_*(\omega_f \otimes X) \to f_{(1)}X$ and hence by the Yoneda lemma we are done.

(4.17) This is the right adjoint of (4.16).

(4.18) This follows by taking $\mathbb{1}_{\mathsf{T}}$ in (4.17) as f^* is strong monoidal.

(4.19) We have isomorphisms

$$X \otimes f_{(1)}Y \simeq X \otimes f_*(\omega_f \otimes Y) \simeq f_*(f^*X \otimes \omega_f \otimes Y) \simeq f_{(1)}(f^*X \otimes Y)$$

by using the ur-Wirthmüller isomorphism (4.16), the projection formula Proposition 3.6 and again the ur-Wirthmüller isomorphism in turn.

(4.20) Fix X in (4.19) and take right adjoints.

(4.21) Fix Y in (4.19) and take right adjoints.

□

Remark 4.23. We have proved that Grothendieck–Neeman duality implies a whole host of isomorphisms as listed in Corollary 4.10. However, we have some converses too. For example, (4.12) implies Grothendieck–Neeman duality since it ensures that $f^{(1)}$ preserves coproducts. We leave it as an exercise (Exercise A.11) for the reader to work out which other isomorphisms listed there imply Grothendieck–Neeman duality.

Remark 4.24. One may wonder why the rather odd name of the *ur*-Wirthmüller isomorphism. The prefix "ur-" means primitive or original, and indeed, later on we will see the Wirthmüller isomorphism. The difference boils down to the fact, that in the "ur-" version, we realise $f_{(1)}$ as a twisted form of f_* where in the full blown Wirthmüller isomorphism, we can also realise f_* as a twisted form of $f_{(1)}$. However, the existence of a full Wirthmüller isomorphism requires a stronger setup than Grothendieck–Neeman duality.

Let us return to our running examples. Let's deal with modular representation theory first since there is less to say here:

Example 4.25. Consider the geometric functor $f^* = \mathrm{res}^G_H \colon \mathsf{StMod}(kG) \to \mathsf{StMod}(kH)$. It is easy to see that this preserves products, and hence has Grothendieck–Neeman duality. In this case, the ur-Wirthmüller isomorphism is the well-known fact that induction $kG \otimes_{kH} -$ and coinduction $\mathrm{Hom}_{kH}(kG, -)$ are isomorphic. (It follows in particular that $\omega_f = \mathbb{1}$.)

Remark 4.26. The previous example highlights an important point: the general framework here does not tell you how to work out what ω_f is (although there are some partial results in this direction which we may revisit if there is time). Often giving an explicit description of ω_f relies on some pre-existing knowledge of the case at hand.

Let us now turn to commutative algebra where we already see several interesting features. We first make the following trivial observation:

Lemma 4.27. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. If U is compactly generated by its tensor unit (i.e., $\mathsf{U} = \mathrm{Loc}(\mathbb{1}_{\mathsf{U}})$) then f^* has Grothendieck–Neeman duality if and only if $f_*(\mathbb{1}_{\mathsf{U}})$ is compact.*

Proof. Recall from Proposition 2.12 that $\mathsf{U}^c = \mathrm{thick}(\mathbb{1}_{\mathsf{U}})$. Notice that $\{X \in \mathsf{U} \mid f_*X \text{ is compact}\}$ is a thick subcategory of U. So if $f_*\mathbb{1}_{\mathsf{U}}$ is compact, then f_* preserves compacts. □

Example 4.28. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring (i.e., $\mathfrak{m}$ is the maximal ideal, and $k = R/\mathfrak{m}$ is the residue field). Let $f^* = k \otimes_R - \colon \mathsf{D}(R) \to \mathsf{D}(k)$ be the geometric functor given by extension of scalars along the quotient map $R \to k$. Recall that the right adjoint to this is the restriction of scalars functor. This satisfies Grothendieck–Neeman duality if and only if $k \in \mathsf{D}(R)^c$. By Example 2.13, this is equivalent to k having finite projective dimension over R. It is a standard fact in commutative algebra that this is equivalent to R being a regular local ring (i.e., if $\mathfrak{m} = (x_1, \cdots, x_n)$ with n minimal, then $\dim(R) = n$). We immediately see that this is rather restrictive from a commutative algebraist's perspective: regular local rings form a very small class of the rings we are interested in. As an example to keep in mind for later, $R = k[x]/x^2$ is not regular: it has Krull dimension 0, but the maximal ideal can't be generated by 0 elements! (In fact, fields are the only regular local rings of Krull dimension 0.) This in particular motivates a generalisation of the above results which weaken the requirement that k is compact in $\mathsf{D}(R)$.

5. Proxy-small objects

In this section we introduce a substantial weakening of the notion of compact objects. It is motivated by the observation that for a commutative Noetherian local ring $(R, \mathfrak{m}, k)$ the residue field k is compact if and only if R is regular, which is a very restrictive condition. On the other hand, we will see that k is always proxy-small in $\mathsf{D}(R)$.

5.A. Proxy-small objects.

We saw in Theorem 4.8 that f^* has Grothendieck–Neeman duality if and only if f_* preserves compact. By Example 4.28 this is a rather restrictive assumption in general. However, it is not so rare that $f_*(\mathbb{1}_{\mathsf{U}})$ is "almost" compact in the following sense:

Definition 5.1. Let T be a rigidly-compactly generated tt-category. We say that a set of objects $\mathcal{K}$ of T is *proxy-small* if there exists a set of compact objects $\mathcal{W}$ such that $\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(\mathcal{K})$ and $\mathcal{K} \subseteq \mathrm{Loc}_{\otimes}(\mathcal{W})$. We call $\mathcal{W}$ the *witness* for the proxy-smallness of $\mathcal{K}$, and note this implies $\mathrm{Loc}_{\otimes}(\mathcal{K}) = \mathrm{Loc}_{\otimes}(\mathcal{W})$.

Remark 5.2. It is helpful to introduce some terminology to make discussing proxy-small objects easier. If $X \in \mathrm{thick}_{\otimes}(Y)$, we say that Y *finitely builds* X. If $X \in \mathrm{Loc}_{\otimes}(Y)$ we say that Y *builds* X.

A priori, the definition of proxy-smallness depends on the choice of *some* witness, but one can give an equivalent definition in terms which are intrinsic to $\mathcal{K}$:

Lemma 5.3. *Let $\mathcal{K}$ be a set of objects in a rigidly-compactly generated tt-category T. Then $\mathcal{K}$ is proxy-small if and only if* $\mathrm{Loc}_{\otimes}(\mathcal{K}) = \mathrm{Loc}_{\otimes}(\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c)$.

Proof. By definition, $\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c$ is a set of compact objects contained in $\mathrm{thick}_{\otimes}(\mathcal{K})$. The reverse implication is then immediate since we may take $\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c$ as the witness. For the forward implication, we have a witness $\mathcal{W}$ set which must satisfy $\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c$. Since $\mathcal{K} \subseteq \mathrm{Loc}_{\otimes}(\mathcal{W})$ it follows that $\mathcal{K} \subseteq \mathrm{Loc}_{\otimes}(\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c)$. The other inclusion required is trivial. □

Remark 5.4. Despite the definition being independent of the choice of witness, we note that choosing a witness is often convenient. This is illustrated in particular by some of the following examples.

Example 5.5. Proxy-small objects are a weakening of the notation of compact objects (an unfortunate terminology clash gives rise to the small vs compact: some authors call compact objects small). If $\mathcal{K}$ is a set of compact objects of T, then $\mathcal{K}$ is proxy-small by taking $\mathcal{W} = \mathcal{K}$.

Example 5.6. Let k be a field. Any object M in $\mathsf{D}(k)$ is proxy-small with k as a witness. Indeed, since any object of $\mathsf{D}(k)$ may be written as a sum of shifted copies of k we have $M \simeq \oplus_{i \in I} \Sigma^{a_i} k$. The claim then follows clearly from the definition.

Example 5.7. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring $(R, \mathfrak{m}, k)$. We claim that $k = R/\mathfrak{m}$ is proxy-small with witness the Koszul complex on the generators of $\mathfrak{m}$. In contrast, recall that the residue field k is compact if and only if R is regular. Proving this requires us setting up a bit of background which we do below, see Corollary 5.17.

Example 5.8. We give one more example from commutative algebra. Recall that $(R, \mathfrak{m}, k)$ is a complete intersection if its completion is a regular local ring quotiented by an ideal generated by a regular sequence. Write $\mathsf{D}^f(R)$ for the full subcategory of $\mathsf{D}(R)$ on those complexes whose homology is bounded and degreewise finitely generated. It is a recent theorem of Pollitz (building on work of Dwyer–Greenlees–Iyengar) that a commutative local Noetherian ring $(R, \mathfrak{m}, k)$ is a complete intersection if and only if every object of $\mathsf{D}^f(R)$ is proxy-small.

We leave various closure properties of proxy-small objects as an exercise, see Exercise A.12.

Let us also give a non-example.

Example 5.9. Let $R = k[x_1, x_2, \ldots]$ be the infinite polynomial ring. Then the residue field k is not proxy-small in $\mathsf{D}(R)$. Suppose that k is proxy-small with witness $\mathcal{W}$ and argue by contradiction. Recall that $Dk \simeq 0$ in $\mathsf{D}(R)$. Since $\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(k)$ it follows that $D\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(Dk)$ and hence $DW = 0$ for all $W \in \mathcal{W}$. Since $\mathcal{W}$ is a set of compacts (hence rigids), we have $W \simeq D^2 W$ (see Lemma 2.20), and hence $W = 0$ too. Note that this argument applies more generally in any rigidly-compactly generated tt-category and to any object k with $Dk \simeq 0$. For example, it also applies to show that $\mathbb{F}_p$ is not proxy-small in the stable homotopy category of spectra.

5.B. **Proxy-smallness of residue fields.** We now return to Example 5.7 and justify it. In order to do this, we need to make some recollections on homotopy colimits:

Definition 5.10. Let T be a triangulated category with coproducts, and let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system of maps in T. The *homotopy colimit* of this system, denoted $\mathrm{hocolim} X_i$, is the cone of the map

$$\bigoplus X_i \xrightarrow{1-f} \bigoplus X_i$$

which on the nth component is given by

$$X_n \xrightarrow{x \mapsto (x, -f_n(x))} X_n \oplus X_{n+1} \xrightarrow{\mathrm{inc}} \bigoplus X_i.$$

Lemma 5.11. *Let T be a triangulated category with coproducts, and let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system of maps in T. Let $Y \in \mathsf{T}$, and suppose that there exists maps $g_n \colon X_n \to Y$ such that $g_n = g_{n+1} f_n$ for all n. Then there exists a map $\overline{g} \colon \operatorname{hocolim} X_i \to Y$ such that $g = \overline{g} q$ where q is the induced map $q \colon \bigoplus X_i \to \operatorname{hocolim} X_i$.*

Proof. By universal property of the coproduct, we obtain a map $g \colon \bigoplus X_i \to Y$ by assembling the g_i. Note that $g(1-f) = 0$ since on the nth component we have $g(1-f)(x) = g(x_n, -f_n(x_n)) = g_n(x_n) - g_{n+1} f_n(x_n) = 0$. So consider the diagram

$$\begin{tikzcd}
\bigoplus X_i \arrow[r, "1-f"] \arrow[d] & \bigoplus X_i \arrow[r, "q"] \arrow[d, "g"] & \operatorname{hocolim} X_i \arrow[r] & \Sigma \bigoplus X_i \arrow[d] \\
0 \arrow[r] & Y \arrow[r, "\mathrm{id}"'] & Y \arrow[r] & 0.
\end{tikzcd}$$

The top row is a triangle by definition of the homotopy colimit, and the bottom row is a triangle by (TR0). The left hand square commutes since $g(1-f) = 0$, so by (TR3) there exists a map $\overline{g} \colon \operatorname{hocolim} X_i \to Y$ making the diagram commute. □

Lemma 5.12. *Let T be a triangulated category with coproducts. Let C be a compact object and let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system in T. Then the canonical map*

$$\varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i)$$

is an isomorphism.

Proof. By definition of the homotopy colimit and since $\operatorname{Hom}(C, -)$ is a homological functor we obtain a long exact sequence,

$$\cdots \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus \Sigma X_i) \to \cdots .$$

We therefore have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & 0 \\
\cdots \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \arrow[r] & \cdots
\end{tikzcd}$$

The first two columns are isomorphisms since C is compact, and so we see that the map $\operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i)$ is injective. Similarly, we may apply the same argument to ΣX_i to deduce that $\operatorname{Hom}_{\mathsf{T}}(C, \bigoplus \Sigma X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus \Sigma X_i)$ is also injective. Therefore the long exact sequence collapses into short exact sequences. We then have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \arrow[r] & 0
\end{tikzcd}$$

in which the first two columns are isomorphisms as C is compact. The result then follows. □

Corollary 5.13. *Let T be a compactly generated triangulated category. Let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system in T, and let $Y \in \mathsf{T}$ such that there exists maps $g_n \colon X_n \to Y$ with $g_n = g_{n+1} f_n$ for all n. If the canonical map*

$$\varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, Y)$$

is an isomorphism for all $C \in \mathsf{T}^{\mathrm{c}}$, then $Y \simeq \operatorname{hocolim} X_i$.

Proof. By Lemma 5.11 we obtain a map $\overline{g}\colon \operatorname{hocolim} X_i \to Y$. Since T is compactly generated, to prove this is an isomorphism it suffices to check that the induced map

$$\operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, Y)$$

is an isomorphism for all $C \in \mathsf{T}^{\mathrm{c}}$. This is immediate from Lemma 5.12. □

We now have the necessary background on homotopy colimits to return to the example of commutative Noetherian rings.

Definition 5.14. Let R be a commutative ring and $I = (x_1, \cdots, x_n)$ be a finitely generated ideal.

(1) The *unstable Koszul complex* $K(I)$ is defined by

$$K(I) = \bigotimes_{i=1}^{n} (R \xrightarrow{\cdot x} R)$$

where the complex $R \xrightarrow{\cdot x} R$ is in degrees 0 and -1. Note that by definition, this is a compact object in $\mathsf{D}(R)$.

(2) The *stable Koszul complex* $K_\infty(I)$ is defined by

$$K_\infty(I) = \bigotimes_{i=1}^{n} (R \to R[1/x_i])$$

where the complex $R \to R[1/x]$ is in degrees 0 and -1.

For an element $x \in R$, we note that we have a map $K(x^i) \to K(x^{i+1})$ for all i, represented by the commutative diagram

$$\begin{tikzcd} R \arrow[r, "\cdot x^i"] \arrow[d, "1"'] & R \arrow[d, "\cdot x"] \\ R \arrow[r, "\cdot x^{i+1}"'] & R \end{tikzcd}$$

Therefore we have a tower $\cdots \to K(x^i) \to K(x^{i+1}) \to \cdots$ and we may take the homotopy colimit of this.

Lemma 5.15. *Let $x \in R$. Then* $\operatorname{hocolim}(K(x^i)) \simeq K_\infty(x)$.

Proof. By Corollary 5.13 it suffices to show that we have compatible maps $K(x^i) \to K_\infty(x)$ for all i, so that the induced map

$$\varinjlim \operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^j R, K(x^i)) \to \operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^j R, K_\infty(x))$$

is an isomorphism for all j. The existence of compatible maps is clear from the commutative diagram

$$\begin{tikzcd} R \arrow[r, "\cdot x^i"] \arrow[d] & R \arrow[d] \\ R \arrow[r] & R[1/x]. \end{tikzcd}$$

Recall that $\operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^j R, -) = H_j(-)$, so it suffices to treat the cases $j = 0$ and $j = -1$. For $j = -1$, $H_{-1}(K(x^i)) = R/x^i$ and $H_{-1}(K_\infty(x)) = R[1/x]/R = R/x^\infty$. Therefore we see that the claim holds for $j = -1$. For $j = 0$, $H_0(K(x^i)) = \{m \mid x^i m = 0\}$ and $H_0(K_\infty(x)) = \{m \mid x^i m = 0 \text{ for some } i\}$. As such we see that $\varinjlim H_0(K(x^i)) = H_0(K_\infty(x))$ which completes the proof of the claim. □

Theorem 5.16. *Let R be a commutative ring and $I = (x_1, \cdots, x_n)$ be a finitely generated ideal. Then*

$$\mathrm{Loc}(R/I) = \mathrm{Loc}(K(I)) = \mathrm{Loc}(K_\infty(I)).$$

If R/I is moreover a regular ring, then $K(I) \in \mathrm{thick}(R/I)$.

Proof. We will prove that (1) R/I builds $K(I)$, (2) $K(I)$ builds $K_\infty(I)$, and (3) $K_\infty(I)$ builds R/I. Note that this suffices since it implies that

$$\mathrm{Loc}(R/I) \subseteq \mathrm{Loc}(K_\infty(I)) \subseteq \mathrm{Loc}(K(I)) \subseteq \mathrm{Loc}(R/I).$$

(1) For each $i \in \mathbb{Z}$ there is a triangle

$$H_i(K(I)) \to \tau_{\leq i} K(I) \to \tau_{\leq i-1} K(I)$$

where $\tau_{\leq i}$ denotes the smart truncation (i.e., $(\tau_{\leq i} X)_j = X_j$ for $j < i$, is 0 for $j > i$ and is $\mathrm{coker}(d_{j+1})$ for $j = i$). We now claim that the homology groups of $K(I)$ are modules over R/I. To see this, note that multiplication by x is null homotopic as a map $K(x) \to K(x)$. As such, for any $x \in I$, $x\colon K(I) \to K(I)$ is null homotopic. Therefore x acts as 0 in homology, and as such the homology groups of $K(I)$ are all R/I-modules. Therefore $H_i(K(I))$ may be built from R/I in $\mathsf{D}(R/I)$. The restriction of scalars functor $\mathsf{D}(R/I) \to \mathsf{D}(R)$ is triangulated and preserves coproducts, and hence R/I also builds $H_i(K(I))$ in $\mathsf{D}(R)$. Now we argue by induction that $\tau_{\leq i} K(I)$ is built from R/I, for each $-n \leq i \leq 0$. We start with $i = -n$ as the base case. When $i = -n$, $\tau_{\leq i-1} K(I) \simeq 0$ so the base case holds by the above triangle. Then inducting, we obtain that $\tau_{\leq i} K(I)$ is built from R/I for each $-n \leq i \leq 0$. When $i = 0$, $\tau_{\leq i} K(I) = K(I)$ so this gives the claim as desired. If R/I is moreover regular, then note that $H_i(K(I))$ is finitely built from R/I in $\mathsf{D}(R/I)$, so the same proof then shows that R/I finitely builds $K(I)$ in $\mathsf{D}(R)$.

(2) One can check that there is a triangle $K(x^{i-1}) \to K(x^i) \to K(x)$, and so by induction we have that $K(x^i) \in \mathrm{thick}(K(x))$ for all i. Since $K_\infty(x)$ is the homotopy colimit of the $K(x^i)$ by Lemma 5.15, it follows that $K_\infty(x)$ lies in the localizing subcategory generated by $K(x)$. The general case follows from this by tensoring.

(3) The stable Koszul complex $K_\infty(I)$ is a dg-flat complex, and hence $R/I \otimes^{\mathsf{L}}_R K_\infty(I) = R/I \otimes_R K_\infty(I)$. As $R/I \otimes_R R[1/x_i] \simeq 0$, one sees that $R/I \otimes^{\mathsf{L}}_R K_\infty(I) \simeq R/I$. As R builds any M in $\mathsf{D}(R)$, $K_\infty(I) \simeq K_\infty(I) \otimes^{\mathsf{L}}_R R$ builds $K_\infty(I) \otimes^{\mathsf{L}}_R M$ for all $M \in \mathsf{D}(R)$. Therefore $K_\infty(I)$ builds $R/I \otimes^{\mathsf{L}}_R K_\infty(I) \simeq R/I$ as required.

Therefore we have the above chain of inclusions of localising subcategories, and hence they are all equal. □

Corollary 5.17. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring. Then k is proxy-small in $\mathsf{D}(R)$.*

Proof. Applying Theorem 5.16 to $I = \mathfrak{m}$ we see that $K(\mathfrak{m}) \in \mathrm{thick}(k)$ and $k \in \mathrm{Loc}(K(\mathfrak{m}))$. Since $K(\mathfrak{m})$ is also compact by definition it follows that k is proxy-small in $\mathsf{D}(R)$ with witness $K(\mathfrak{m})$. □

6. Torsion and completion

6.A. **Localisation theory.** In this subsection we set up the basic theory of localisation in triangulated and tt-categories. We start off in just the triangulated setting.

Definition 6.1. A triangulated functor $L\colon \mathsf{T} \to \mathcal{S}$ is a *localization* if it has a fully faithful right adjoint. Note then that $\mathcal{S} \simeq L\mathsf{T}$, the essential image of L. Frequently we write L for the composite of L followed by the inclusion. A *colocalization* is the dual concept.

Remark 6.2. Some authors might call the above a *Bousfield localization* to distinguish it from other types of localizations. In this course, we will only see Bousfield localizations.

Remark 6.3. An alternative more explicit definition of a localization is as follows. A localization is an exact functor $L\colon \mathsf{T} \to \mathsf{T}$ together with a natural transformation $\eta\colon \mathrm{id} \to L$ such that $L\eta\colon L \to L^2$ is an isomorphism, and $L\eta = \eta L$. Note the difference between the codomains in the definitions.

Definition 6.4. Let L be a localization on T. An object $X \in \mathsf{T}$ is said to be L-local if the unit map $X \to LX$ is an isomorphism, and is said to be L-acyclic if $LX \simeq 0$.

With the definitions set, we may now turn to proving various properties of (co)localizations. Firstly, we show that localizations and colocalizations always come in pairs.

Proposition 6.5. *Let T be a triangulated category. There is a one-to-one correspondence between localizations of T and colocalizations of T, in which L and Γ correspond if and only if $\Gamma X \to X \to LX \to \Sigma\Gamma X$ is a triangle for all $X \in \mathsf{T}$.*

Proof. Suppose that L is a localization of T. By (TR1) we may extend the unit map $X \to LX$ to give a triangle

$$\Gamma X \to X \to LX \to \Sigma\Gamma X.$$

By applying the triangulated functor L and using that $L \xrightarrow{\sim} L^2$ is an isomorphism we see that $L\Gamma X \simeq 0$, so ΓX is L-acyclic. Conversely, suppose that Z is L-acyclic. Then $\mathrm{Hom}_{\mathsf{T}}(Z, LX) = 0$ since $\mathrm{Hom}_{\mathsf{T}}(Z, LX) = \mathrm{Hom}_{\mathsf{T}}(LZ, LX) = 0$ by adjunction. Therefore, applying $\mathrm{Hom}_{\mathsf{T}}(Z, -)$ to the above triangle, we see that $\mathrm{Hom}_{\mathsf{T}}(Z, \Gamma X) = \mathrm{Hom}_{\mathsf{T}}(Z, X)$. Therefore ΓX has a universal property: if Z is L-acyclic and there is a map $f\colon Z \to X$, then f factors as $Z \to \Gamma X \to X$.

Normally applying (TR3) to construct maps is not functorial, but the above universal property of ΓX allows us to make Γ functorial as we now describe. Let $f\colon A \to X$ be a morphism in T. By (TR3) we obtain a map $\Gamma(f)$ in the commutative diagram

$$\begin{tikzcd}
\Sigma^{-1}LA \arrow[r] \arrow[d] & \Gamma A \arrow[r] \arrow[d, "\Gamma(f)"] & A \arrow[r] \arrow[d] & LA \arrow[d] \\
\Sigma^{-1}LX \arrow[r] & \Gamma X \arrow[r] & X \arrow[r] & LX
\end{tikzcd}$$

So to see that Γ is functorial, it suffices to show that $\Gamma(f)$ is unique which is immediate from the universal property. Therefore, given a localization L, we have constructed a functor $\Gamma\colon \mathsf{T} \to \mathsf{T}$.

It remains to prove that Γ is triangulated and is a colocalization. Firstly, we consider the diagram

$$\begin{tikzcd}
LX \arrow[r] \arrow[d, "\sim"] & \Sigma\Gamma X \arrow[r] \arrow[d, dashed] & \Sigma X \arrow[r] \arrow[d, "1"] & \Sigma LX \arrow[d, "\sim"] \\
\Sigma^{-1}L\Sigma X \arrow[r] & \Gamma\Sigma X \arrow[r] & \Sigma X \arrow[r] & L\Sigma X
\end{tikzcd}$$

in which the top row is the triangle obtained by applying Σ to the standard triangle, and the bottom row is the standard triangle on ΣX. Using (TR3) and the 5-lemma, we deduce that $\Gamma\Sigma \simeq \Sigma\Gamma$. We use this isomorphism implicitly in what follows.

Consider a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$. We may complete the map Γf to a triangle by (TR1). Consider the diagram

$$\begin{tikzcd}
\Gamma X \arrow[r, "\Gamma f"] \arrow[d, "\epsilon_X"] & \Gamma Y \arrow[r, "\alpha"] \arrow[d, "\epsilon_Y"] & C \arrow[r, "\beta"] \arrow[d, dashed, "\epsilon"] & \Sigma\Gamma X \arrow[d, "\Sigma\epsilon_X"] \\
X \arrow[r, "f"'] & Y \arrow[r, "g"'] & Z \arrow[r, "h"'] & \Sigma X
\end{tikzcd}$$

in which the dashed map exists by (TR3). Suppose that W is L-acyclic. By applying $\mathrm{Hom}_{\mathsf{T}}(W,-)$ and the 5-lemma, we see that $\mathrm{Hom}_{\mathsf{T}}(W,C) \to \mathrm{Hom}_{\mathsf{T}}(W,Z)$ is an isomorphism. Therefore C is terminal amongst L-acyclic objects mapping to Z. We proved above that ΓZ admits the same universal property, and hence $C \simeq \Gamma Z$. As in the proof of functoriality above, one then identifies α and β with Γg and Γh. Therefore Γ is triangulated as required.

It remains to prove that Γ is a colocalization. Consider the inclusion of the L-acyclic objects into T. We have seen that for any L-acyclic Z, we have that $\mathrm{Hom}_{\mathsf{T}}(Z,\Gamma X) \to \mathrm{Hom}_{\mathsf{T}}(Z,X)$ is an isomorphism. Therefore Γ is right adjoint to this inclusion functor. To see that the inclusion is fully faithful, we just need to see that the counit is an isomorphism, i.e., that the map $\epsilon\colon \Gamma Z \to Z$ is an isomorphism if Z is L-acyclic. This is immediate by the standard triangle. □

Next, we give various equivalent characterisations of local objects. We leave the dual colocal version to the reader. We use the following notation: for any subcategory $\mathcal{X}$,

$$\mathcal{X}^{\perp} = \{Y \in \mathsf{T} \mid \mathrm{Hom}_{\mathsf{T}}(X,Y) = 0 \text{ for all } X \in \mathcal{X}\}.$$

Proposition 6.6. *Let T be a triangulated category and L be a localization. Write Γ for corresponding colocalization as in Proposition 6.5. The following are equivalent for an object $X \in \mathsf{T}$:*

(1) X is L-local;
(2) $X \simeq LY$ for some $Y \in \mathsf{T}$;
(3) $\mathrm{Hom}_{\mathsf{T}}(Z,X) \simeq 0$ for all L-acyclic objects Z;
(4) X is Γ-acyclic.

In particular, $\mathrm{im}(L) = \ker(L)^{\perp}$.

Proof. The implication (1) ⇒ (2) is trivial. For (2) ⇒ (3) we have $\mathrm{Hom}_{\mathsf{T}}(Z,X) = \mathrm{Hom}_{\mathsf{T}}(Z,LY) = \mathrm{Hom}_{\mathsf{T}}(LZ,LY)$ since L is a localization. Now this is zero for all L-acyclic Z as required. For (3) ⇒ (4), as $L\Gamma \simeq 0$ as proved in Proposition 6.5, the map $\epsilon\colon \Gamma X \to X$ is zero by assumption, and hence $\Gamma\epsilon = 0$. However, $\Gamma\epsilon\colon \Gamma^2 X \to \Gamma X$ is an isomorphism. Therefore $\Gamma X \simeq 0$ as required. Finally, the implication (4) ⇒ (1) follows immediately from the defining triangle $\Gamma X \to X \to LX \to \Sigma\Gamma X$. □

Corollary 6.7. *Let L be a localization. Then the full subcategory $\ker(L)$ of L-acyclics is localizing, and the full subcategory $\mathrm{im}(L)$ of L-local objects is colocalizing.*

Proof. Since L is a triangulated functor, $\ker(L)$ is thick. Now $X \in \ker(L)$ if and only if $\mathrm{Hom}_{\mathsf{T}}(X,Y) = 0$ for all L-local objects Y by the dual form of Proposition 6.6. Therefore $\ker(L)$ is closed under coproducts, and hence is localizing. The argument that $\mathrm{im}(L)$ is colocalizing is similar. □

We now turn to consider localization functors on tensor-triangulated categories. In order to ensure good behaviour with the tensor product, it is necessary to refine the notion of localization slightly.

Definition 6.8. A localization functor $L\colon \mathsf{T} \to \mathsf{T}$ is said to be *monoidal* if $LX \simeq 0$ implies that $L(X \otimes Y) \simeq 0$ for all $Y \in \mathsf{T}$. A colocalization functor $\Gamma\colon \mathsf{T} \to \mathsf{T}$ is *monoidal* if $\Gamma X \simeq 0$ implies that $\Gamma\underline{\mathrm{Hom}}(Y,X) \simeq 0$ for all $Y \in \mathsf{T}$.

Remark 6.9. Notice that if Γ is monoidal, then $\mathrm{im}(\Gamma)$ is a localising $\otimes$-ideal. If $X \in \mathrm{im}(\Gamma) = \ker(L)$ and $Y \in \mathsf{T}$, then $L(X \otimes Y) \simeq 0$ since L is monoidal (see Exercise A.13), so $X \otimes Y \in \mathrm{im}(\Gamma)$ as claimed. We leave it to the reader to state and prove the analogous statement for L.

Lemma 6.10. *Let $L\colon \mathsf{T} \to \mathsf{T}$ be a monoidal localization. There is a natural map $\alpha_X\colon L\mathbb{1} \otimes X \to LX$ for all $X \in \mathsf{T}$.*

Proof. Applying L to the defining triangle we have a triangle

$$L(\Gamma\mathbb{1} \otimes X) \to LX \to L(L\mathbb{1} \otimes X) \to \Sigma L(\Gamma\mathbb{1} \otimes X).$$

The first entry is 0 since L is monoidal, and hence $LX \to L(L\mathbb{1} \otimes X)$ is an isomorphism. Therefore we have a natural map $L\mathbb{1} \otimes X \to L(L\mathbb{1} \otimes X) \simeq LX$ as required. □

Proposition 6.11. *Let T be a tensor-triangulated category, and let L be a monoidal localization. Write $\mathcal{S}$ for the full subcategory of L-local objects. The following are equivalent:*

(i) The natural map $\alpha_X \colon L\mathbb{1} \otimes X \to LX$ from Lemma 6.10 is an equivalence for all $X \in \mathsf{T}$.
(ii) $i \colon \mathcal{S} \to \mathsf{T}$ preserves coproducts (equivalently, $L \colon \mathsf{T} \to \mathsf{T}$ preserves coproducts).
(iii) $\mathcal{S}$ is a localizing subcategory of T.

If any of these equivalent conditions hold, we say that L is smashing.

Proof. This is left as Exercise A.15. □

Proposition 6.12. *Let $L \colon \mathsf{T} \to \mathsf{T}$ be a monoidal localisation functor. Then the category $L\mathsf{T}$ of L-local objects is a tensor-triangulated category with:*

- *triangles are those $X \to Y \to Z$ which are triangles in T;*
- *monoidal product $X \otimes_L Y \coloneqq L(X \otimes Y)$;*
- *tensor unit $\mathbb{1}_{L\mathsf{T}} = L\mathbb{1}_{\mathsf{T}}$;*
- *internal hom $\underline{\mathrm{Hom}}_L(X, Y) \coloneqq \underline{\mathrm{Hom}}(X, Y)$.*

The localisation functor $L \colon \mathsf{T} \to L\mathsf{T}$ is then a coproduct preserving tt-functor.

Proof. Since $L\mathsf{T}$ is a thick subcategory of T, the above makes $L\mathsf{T}$ into a triangulated category. It is clear that the monoidal product and tensor unit defined above are L-local. Notice that the internal hom is automatically L-local: by Proposition 6.6 it suffices to show that $\mathrm{Hom}(Z, \underline{\mathrm{Hom}}(X, Y)) = 0$ if $LZ \simeq 0$. By adjunction we have

$$\mathrm{Hom}(Z, \underline{\mathrm{Hom}}(X, Y)) \simeq \mathrm{Hom}(Z \otimes X, Y) \simeq 0$$

where the final isomorphism holds by definition of a monoidal localisation. It is then a straightforward check that the tensor product and internal hom are adjoint triangulated functors. □

Remark 6.13. Notice that if L is a smashing localisation then it is not necessary to apply L to obtain the correct tensor product since $L(X \otimes Y) = L\mathbb{1} \otimes X \otimes Y = X \otimes Y$ since X (or/and Y) is L-local.

Remark 6.14. Notice that if T is rigidly-compactly generated, then $L\mathsf{T}$ need not be rigidly-compactly generated. It is left as an exercise (see Exercise A.16) to prove that $L\mathsf{T}$ is rigidly-compactly generated if and only if L is smashing.

We now have the required background to prove the existence of (co)localisation functors based on sets of compact objects.

Theorem 6.15. *Let T be a tt-category which is compactly generated by rigid objects and let $\mathcal{K} \subseteq \mathsf{T}^c$. Define*

$$\mathsf{T}^{\mathcal{K}-\mathrm{tors}} \coloneqq \mathrm{Loc}_{\otimes}(\mathcal{K}), \qquad \mathsf{T}^{\mathcal{K}-\mathrm{loc}} \coloneqq (\mathsf{T}^{\mathcal{K}-\mathrm{tors}})^{\perp}, \qquad \mathsf{T}^{\mathcal{K}-\mathrm{comp}} \coloneqq (\mathsf{T}^{\mathcal{K}-\mathrm{loc}})^{\perp}.$$

There are monoidal localization and colocalization functors:

$$\Gamma \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-tors}}, \quad L \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-loc}}, \quad V \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-loc}}, \quad \Lambda \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-comp}}$$

fitting into triangles $\Gamma X \to X \to LX$ and $VX \to X \to \Lambda X$. These satisfy the following properties:

(1) We have $\ker(L) = \mathrm{im}(\Gamma) = \mathsf{T}^{\mathcal{K}\text{-tors}}$;

(2) We have $\ker(\Gamma) = \operatorname{im}(L) = \operatorname{im}(V) = \ker(\Lambda) = \mathsf{T}^{\mathcal{K}\text{-loc}}$*;*
(3) We have $\ker(V) = \operatorname{im}(\Lambda) = \mathsf{T}^{\mathcal{K}\text{-comp}}$*.*
(4) for any $X \in \mathsf{T}$*,* $LX \simeq L\mathbb{1} \otimes X$ *and* $\Gamma X \simeq \Gamma\mathbb{1} \otimes X$*;*
(5) for all $X \in \mathsf{T}$*, the natural maps* $\Gamma X \to \Gamma\Lambda X$ *and* $\Lambda\Gamma X \to \Lambda X$ *are isomorphisms;*
(6) the functors Γ *and* Λ *give an equivalence of categories* $\mathsf{T}^{\mathcal{K}\text{-tors}} \simeq \mathsf{T}^{\mathcal{K}\text{-comp}}$*.*

Proof. Writing $\mathcal{G}$ for a set of compact-rigid generators for T, since $\operatorname{Loc}_{\otimes}(\mathcal{K}) = \operatorname{Loc}(\mathcal{K} \otimes \mathcal{G})$, the category $\mathsf{T}^{\mathcal{K}-\text{tors}}$ is compactly generated. Therefore by Theorem 3.2, the inclusion $\mathsf{T}^{\mathcal{K}-\text{tors}} \hookrightarrow \mathsf{T}$ has a right adjoint Γ. This has a corresponding localisation functor L. We define $V = \underline{\operatorname{Hom}}(L\mathbb{1}, -)$ and $\Lambda = \underline{\operatorname{Hom}}(\Gamma\mathbb{1}, -)$. These are triangulated functors. We will verify that these are (co)localizations later on.

Proof of (1): By construction, $\operatorname{im}(\Gamma) = \mathsf{T}^{\mathcal{K}-\text{tors}}$. Note that $\ker(L)$ is localizing by Corollary 6.7 and contains $\mathcal{K} \otimes \mathcal{G}$. Therefore $\mathsf{Loc}^{\otimes}(\mathcal{K}) \subseteq \ker(L)$. Conversely, if $LX \simeq 0$, then $X \simeq \Gamma X \in \mathsf{Loc}^{\otimes}(\mathcal{K})$.

Proof of (4): Note that $\Gamma X \simeq 0$ if and only if $\operatorname{Hom}_{\mathsf{T}}(Z, X) = 0$ for all $Z \in \operatorname{Loc}_{\otimes}(\mathcal{K})$ as $\operatorname{Hom}_{\mathsf{T}}(Z, X) = \operatorname{Hom}_{\mathsf{T}}(Z, \Gamma X)$. So if $\Gamma X \simeq 0$, then $\operatorname{Hom}_{\mathsf{T}}(Z, \underline{\operatorname{Hom}}(Y, X)) = \operatorname{Hom}_{\mathsf{T}}(Y, \underline{\operatorname{Hom}}(Z, X)) = 0$ and hence $\Gamma\underline{\operatorname{Hom}}(Y, X) \simeq 0$ for all $Y \in \mathsf{T}$ so that Γ is monoidal. By Exercise A.13, it follows that L is monoidal. Now X is L-local if and only if $\operatorname{Hom}_{\mathsf{T}}(Z, X) = 0$ for all $Z \in \operatorname{thick}_{\otimes}(\mathcal{K})$. Since any such Z is compact, we see that the category of L-local objects is closed under coproducts, and hence L is smashing by Proposition 6.11. By the 5 lemma, it follows that $\Gamma X \simeq \Gamma\mathbb{1} \otimes X$ for all $X \in \mathsf{T}$.

Proof of (2): Notice that (4) shows that V and Λ are idempotent functors. We have $\operatorname{im}(L) = \ker(L)^{\perp} = \ker(\Gamma) = \mathsf{T}^{\mathcal{K}\text{-loc}}$ by Proposition 6.6 together with (1). By the idempotence of V together with the associated triangle $VX \to X \to \Lambda X$, we also see that $\ker(\Lambda) = \operatorname{im}(V)$. So it remains to prove that $\ker(\Lambda) = \mathsf{T}^{\mathcal{K}\text{-loc}}$.

If $X \in \mathsf{T}^{\mathcal{K}\text{-loc}}$ then $\operatorname{Hom}(\Gamma\mathbb{1} \otimes G, X) \simeq 0$ for all $G \in \mathcal{G}$ by (1). Therefore $\Lambda\mathbb{1} \simeq \underline{\operatorname{Hom}}(\Gamma\mathbb{1}, X) \simeq 0$. Conversely, suppose that $\Lambda X \simeq 0$ and $Y \in \mathsf{Loc}^{\otimes}(\mathcal{K})$. Then $\operatorname{Hom}_{\mathsf{T}}(Y, X) = \operatorname{Hom}_{\mathsf{T}}(\Gamma Y, X)$ by (1). Since Γ is smashing, this is equal to $\operatorname{Hom}_{\mathsf{T}}(\Gamma\mathbb{1} \otimes Y, X) \simeq \operatorname{Hom}_{\mathsf{T}}(Y, \Lambda X) \simeq 0$.

Proof of (3): This follows the same strategy as the second paragraph of (2).

Proof of (5): We have $\Lambda LX \simeq 0$ for all X by (2). Therefore by applying Λ to the triangle $\Gamma X \to X \to LX$ we conclude that $\Lambda\Gamma X \to \Lambda X$ is an isomorphism. The other isomorphism is similar.

Proof of (6): This is immediate from (1), (3), and (5).

Finally we argue that Λ is a localization functor. To do this, it suffices to show that the canonical map $\operatorname{Hom}_{\mathsf{T}}(\Lambda X, \Lambda Y) \to \operatorname{Hom}_{\mathsf{T}}(X, \Lambda Y)$ is an equivalence. This is equivalent to $\operatorname{Hom}_{\mathsf{T}}(VX, \Lambda Y) \simeq 0$. Now by (2), $VX \in \mathsf{T}^{\mathcal{K}\text{-loc}}$, and by (3) $\Lambda X \in \mathsf{T}^{\mathcal{K}\text{-comp}}$ so the claim follows. Finally, we need that show that Λ is moreover a monoidal localization. To see this, if $\Lambda X \simeq 0$, then $\Gamma X \simeq 0$ by (5). Again by (5) together with (4), $\Lambda(X \otimes Y) \simeq \Lambda\Gamma(X \otimes Y) \simeq \Lambda(\Gamma X \otimes Y) \simeq 0$ as required. □

Remark 6.16. If $\mathcal{K}$ is proxy-small in T, then the previous theorem applies too, since we have a set of compact objects $\mathcal{W}$ such that $\operatorname{Loc}_{\otimes}(\mathcal{K}) = \operatorname{Loc}_{\otimes}(\mathcal{W})$.

Remark 6.17. If confusion may occur, we record the subscript of the set of compact objects on the associated functors, e.g., $\Gamma_{\mathcal{K}}$.

Corollary 6.18 (Greenlees-May duality)**.** *Let* T *be a tt-category which is compactly generated by rigid objects and let* $\mathcal{K} \subseteq \mathsf{T}^c$*. Then*

$$\Gamma\colon \mathsf{T} \rightleftarrows \mathsf{T}\colon \Lambda$$

form an adjoint pair which internalises, i.e., for all $X, Y \in \mathsf{T}$*,*

$$\underline{\operatorname{Hom}}(\Gamma X, Y) \simeq \underline{\operatorname{Hom}}(X, \Lambda Y).$$

Proof. By definition of Λ as in the proof of Theorem 6.15 we have $\Lambda = \underline{\mathrm{Hom}}(\Gamma\mathbb{1}, -)$. Therefore,

$$\underline{\mathrm{Hom}}(\Gamma X, Y) \simeq \underline{\mathrm{Hom}}(\Gamma\mathbb{1} \otimes X, Y) \simeq \underline{\mathrm{Hom}}(X, \underline{\mathrm{Hom}}(\Gamma\mathbb{1}, Y)) \simeq \underline{\mathrm{Hom}}(X, \Lambda Y)$$

since Γ is smashing. Applying $\mathrm{Hom}_{\mathsf{T}}(\mathbb{1}, -)$ to the internalised adjunction yields the external form of the adjunction $\mathrm{Hom}_{\mathsf{T}}(\Gamma X, Y) \simeq \mathrm{Hom}_{\mathsf{T}}(X, \Lambda Y)$. □

Remark 6.19. By Proposition 6.12, the category $\mathsf{T}^{\mathcal{K}\text{-comp}}$ is a tt-category with tensor product given by $X \widehat{\otimes} Y := \Lambda(X \otimes Y)$, tensor unit $\Lambda\mathbb{1}_{\mathsf{T}}$ and internal hom the restriction of the internal hom in T. Using the MGM equivalence, this induces a tt-structure on $\mathsf{T}^{\mathcal{K}\text{-tors}}$ with tensor product given by the underlying tensor product in T, tensor unit $\Gamma\mathbb{1}_{\mathsf{T}}$ and internal hom $\Gamma\underline{\mathrm{Hom}}(-,-)$. The functors $\Gamma\colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-tors}}$ and $\Lambda\colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-comp}}$ are both strong symmetric monoidal functors. Moreover the equivalence of categories $\mathsf{T}^{\mathcal{K}\text{-tors}} \simeq \mathsf{T}^{\mathcal{K}\text{-comp}}$ is an equivalence of tt-categories.

6.B. **Relative proxy-smallness.** We now mix the notion of proxy-smallness with the definition of geometric functors:

Definition 6.20. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. We say that f^* is *proxy-small* if there exists a set of compact objects $\mathcal{W} \subseteq \mathsf{T}^c$ such that:

(1) $f_*\mathbb{1}_{\mathsf{U}}$ is proxy-small with witness $\mathcal{W}$;
(2) $\mathbb{1}_{\mathsf{U}}$ is proxy-small with witness $f^*\mathcal{W}$.

Notice that given (1), since f^* preserves compact objects, condition (2) is just the statement that $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$.

Remark 6.21. Condition (1) is the most important: it turns out that if f^* is *enhanced* then condition (2) is automatic. We will see later why we impose the compatibility condition (2): it ensures that the colocalisation $\Gamma\colon \mathsf{T} \to \mathrm{Loc}_{\otimes}(\mathcal{K})$ behaves well with respect to f^*, namely $f^* \simeq f^*\Gamma$.

Example 6.22. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring and let $f^* = k \otimes^{\mathsf{L}}_R - \colon \mathsf{D}(R) \to \mathsf{D}(k)$. Then f^* is a proxy-small geometric functor by Example 5.7: that gives condition (1), and condition (2) follows since $k \otimes^{\mathsf{L}}_R K(\mathfrak{m})$ is a sum of shifted copies of k.

As with the proxy-small condition on objects (recall Lemma 5.3), the proxy-small condition on geometric functors is independent of the choice of witness.

Lemma 6.23. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor where $f_*\mathbb{1}_{\mathsf{U}}$ is proxy-small in T. Then f^* is proxy-small if and only if $\mathrm{Loc}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}}) = \mathsf{U}$.*

Proof. For the reverse direction, as noted above it suffices to show that $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$. Applying f^* to the definition of $f_*\mathbb{1}_{\mathsf{U}}$ being proxy-small we have that $f^*f_*\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$. By assumption, $f^*f_*\mathbb{1}_{\mathsf{U}}$ builds $\mathbb{1}_{\mathsf{U}}$ so we obtain that $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$ as required. For the forward implication, notice that we have $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$ and $f^*\mathcal{W} \in \mathrm{thick}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}})$. So $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}})$, from which it follows that $\mathrm{Loc}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}}) = \mathsf{U}$. □

Definition 6.24. Let f^* be a proxy-small geometric functor. By applying Theorem 6.15 to T and $\mathcal{W}$ (a witness for the proxy-smallness of $f_*\mathbb{1}_{\mathsf{U}}$), we obtain torsion, localisation and completion functors which we denote by Γ_f, L_f, and Λ_f respectively. Likewise we denote the associated (co)localised categories by $\Gamma_f\mathsf{T}$ and so on.

The relevance of the proxy-small condition on geometric functors is that it ensures good compatibility between f^* and the associated torsion and completion functors Γ_f and Λ_f.

Proposition 6.25. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor. There are natural isomorphisms:*

(1) $\Gamma_f f_* Y \simeq f_* Y$ *for all* $Y \in \mathsf{U}$*;*
(2) $f^* \Gamma_f X \simeq f^* X$ *for all* $X \in \mathsf{T}$*;*
(3) $f^{(1)} \Lambda_f X \simeq f^{(1)} X$ *for all* $X \in \mathsf{T}$*;*
(4) $\Lambda_f f_* Y \simeq f_* Y$ *for all* $Y \in \mathsf{U}$*.*

Proof. For (1), by considering the triangle $\Gamma_f X \to X \to L_f X$ it suffices to show that $f_* Y \in \mathrm{Loc}_\otimes(f_* \mathbb{1}_\mathsf{U})$ as $\ker(L_f) = \mathrm{Loc}_\otimes(f_* \mathbb{1}_\mathsf{U})$. We have $Y \in \mathsf{U} = \mathrm{Loc}_\otimes(f^* f_* \mathbb{1}_\mathsf{U}) = \mathrm{Loc}(f^* f_* \mathbb{1}_\mathsf{U} \otimes u \mid u \in \mathsf{U})$ by proxy-smallness (see Lemma 6.23). By applying the coproduct-preserving triangulated functor f_* we therefore have

$$f_* Y \in \mathrm{Loc}(f_*(f^* f_* \mathbb{1}_\mathsf{U} \otimes u) \mid u \in \mathsf{U}) = \mathrm{Loc}(f_* \mathbb{1}_\mathsf{U} \otimes f_* u \mid u \in \mathsf{U}) \subseteq \mathrm{Loc}_\otimes(f_* \mathbb{1}_\mathsf{U})$$

where the equality uses the projection formula (Proposition 3.6).

For (2), by the Yoneda lemma it suffices to show that the canonical map

$$\mathrm{Hom}(Z, \alpha) \colon \mathrm{Hom}(Z, f^* \Gamma_f X) \to \mathrm{Hom}(Z, f^* X)$$

is an isomorphism for all $Z \in \mathsf{U}$. By proxy-smallness, we have a set $\mathcal{W} \subseteq \mathsf{T}^c$ such that $\mathsf{U} = \mathrm{Loc}_\otimes(f^* \mathcal{W})$. Since $\mathrm{Loc}_\otimes(f^* \mathcal{W}) = \mathrm{Loc}(f^* \mathcal{W} \otimes C \mid C \in \mathsf{U}^c)$ it therefore suffices to show that $\mathrm{Hom}(f^* W \otimes C, \alpha)$ is an isomorphism for all $W \in \mathcal{W}$ and $C \in \mathsf{U}^c$. This is equivalent to $\underline{\mathrm{Hom}}(f^* W, \alpha)$ being an isomorphism for all $W \in \mathcal{W}$ which is in turn equivalent to $f^*(DW) \otimes \alpha$ being an isomorphism for all $W \in \mathcal{W}$ by Lemma 2.23. This now holds since f^* is strong monoidal.

Taking the right adjoint of (1) yields (3), and (4) is the right adjoint to (2). □

Proposition 6.26. *Let* $f^* \colon \mathsf{T} \to \mathsf{U}$ *be a proxy-small geometric functor and write* $\mathcal{W} \subseteq \mathsf{T}^c$ *for a witness. Let* α *be a map in* T*. Then the following are equivalent:*

(1) $W \otimes \alpha$ *is an isomorphism for all* $W \in \mathcal{W}$*;*
(2) $\Gamma_f \alpha$ *is an isomorphism;*
(3) $\Lambda_f \alpha$ *is an isomorphism;*
(4) $f^* \alpha$ *is an isomorphism;*
(5) $f^{(1)} \alpha$ *is an isomorphism.*

Proof. That (1) and (2) are equivalent follows from the facts that Γ_f is smashing and elements of $\mathcal{W}$ are $f_* \mathbb{1}_\mathsf{U}$-torsion. The equivalence between (2) and (3) is then immediate from Theorem 6.15. For (2) $\Rightarrow$ (4), if $\Gamma_f \alpha$ is an isomorphism, then $f^* \alpha \simeq f^* \Gamma_f \alpha$ is an isomorphism by Proposition 6.25. Similarly, (3) $\Rightarrow$ (5) holds since we have $f^! \alpha \simeq f^! \Lambda_f \alpha$ by Proposition 6.25.

To see that (4) $\Rightarrow$ (1), we have $f_* \mathbb{1}_\mathsf{U} \otimes \alpha \simeq f_* f^* \alpha$ by the projection formula (Proposition 3.6). As $\mathcal{W} \subseteq \mathrm{thick}_\otimes(f_* \mathbb{1}_\mathsf{U})$, by a thick subcategory argument it then follows that $W \otimes \alpha$ is an isomorphism for all $W \in \mathcal{W}$. Finally, for (5) $\Rightarrow$ (1), we have

$$\mathrm{Hom}(X, \underline{\mathrm{Hom}}(f_* \mathbb{1}_\mathsf{U}, \alpha)) \simeq \mathrm{Hom}(X, f_* f^! \alpha) \simeq \mathrm{Hom}(f^* X, f^! \alpha)$$

for any $X \in \mathcal{C}$, by (3.9) and adjunction. As such, $\underline{\mathrm{Hom}}(f_* \mathbb{1}_\mathsf{U}, \alpha)$ is an isomorphism. Since $\mathcal{W} \subseteq \mathrm{thick}_\otimes(f_* \mathbb{1}_\mathsf{U})$ we see that $\underline{\mathrm{Hom}}(W, \alpha)$ is an isomorphism for each $W \in \mathcal{W}$. Now since each $W \in \mathcal{W}$ is rigid, W is a retract of $W \otimes DW \otimes W$ (see Exercise A.8). Therefore $W \otimes \alpha$ is a retract of $W \otimes \underline{\mathrm{Hom}}(W, \alpha) \otimes W$, and hence is an isomorphism. □

Remark 6.27. We say that a map α in $\mathcal{C}$ is an f^**-isomorphism* if $f^* \alpha$ is an isomorphism, and similarly for $f^!$, Γ_f, and Λ_f.

6.C. **An aside on enhancements.** In this subsection, we prove the assertion of Remark 6.21; namely that an *enhanced* geometric functor $f^* \colon \mathsf{T} \to \mathsf{U}$ is proxy-small if and only if $f_* \mathbb{1}_\mathsf{U}$ is proxy-small in T.

Let us first make precise what we mean by an enhancement. The key point is that we require a "good" notion of modules. Let T is a tt-category and R is a ring object in T (i.e., we have a multiplication map $\mu\colon R \otimes R \to R$ and a unit map $\eta\colon \mathbb{1} \to R$ satisfying the obvious relations). An R-module M is then an object $M \in \mathsf{T}$ together with an action map $a\colon R \otimes M \to M$ which satisfies the expected relations with μ and η. The problem is that, the category $\mathrm{Mod}_{\mathsf{T}}(R)$ of R-modules in T does not inherit a triangulated structure from T in any obvious way. For our purposes the use of enhancements fixes this issue.

For those "in the know", an *enhanced tt-category* for us means a presentably symmetric monoidal stable ∞-category whose tensor product commutes with colimit separately in each variable. Its homotopy category is then a tt-category in the usual sense, and we say it is rigidly-compactly generated if its homotopy category is so. An *enhanced geometric functor* between rigidly-compactly generated enhanced tt-categories is a colimit-preserving symmetric monoidal functor.

For those not in the know, one can assume that we are working with tt-categories where the category of modules over a ring object is again a tt-category. One does not lose much from just pretending this for the purposes of this aside. As such the following is the key observation:

Proposition 6.28. *Let T be a rigidly-compactly generated enhanced tt-category and let R be a commutative ring object in T. Then the category $\mathrm{Mod}_{\mathsf{T}}(R)$ of R-modules is a rigidly-compactly generated enhanced tt-category.*

Proof. We take for granted that $\mathrm{Mod}_{\mathsf{T}}(R)$ is again an enhanced tt-category, so it suffices to prove that it is rigidly-compactly generated. Note that the tensor product in $\mathrm{Mod}_{\mathsf{T}}(R)$ is $- \otimes_R -$ and so has unit object R. In particular, the free functor $R \otimes -\colon \mathsf{T} \to \mathrm{Mod}_{\mathsf{T}}(R)$ is a strong symmetric monoidal functor with right adjoint the forgetful functor.

Write G for a set of compact generators for T. We claim that $R \otimes \mathsf{G}$ is a set of compact generators for T. By the free-forgetful adjunction,

$$\mathrm{Hom}_{\mathrm{Mod}_{\mathsf{T}}(R)}(R \otimes \mathsf{G}, -) = \mathrm{Hom}_{\mathsf{T}}(\mathsf{G}, -).$$

From this it is immediate that $R \otimes \mathsf{G}$ is a set of compact generators for $\mathrm{Mod}_{\mathsf{T}}(R)$. These are also rigid since $R \otimes -\colon \mathsf{T} \to \mathrm{Mod}_{\mathsf{T}}(R)$ is a strong symmetric monoidal functor, see Lemma 2.23. Therefore we have shown that $\mathrm{Mod}_{\mathsf{T}}(R)$ has a set of rigid-compact generators, and moreover the unit object $\mathbb{1}_{\mathrm{Mod}_{\mathsf{T}}(R)} = R$ is compact. Therefore, $\mathrm{Mod}_{\mathsf{T}}(R)$ is rigidly-compactly generated by Proposition 2.25. □

Recall that a functor $F\colon \mathsf{T} \to \mathsf{U}$ between triangulated categories is *conservative* if $FX \simeq 0$ implies that $X \simeq 0$.

Lemma 6.29. *Let $g^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. The functor $g_*\colon \mathsf{U} \to \mathsf{T}$ is conservative if and only if* $\mathrm{thick}(g^*(\mathsf{T}^c)) = \mathsf{U}^c$.

Proof. For the reverse implication, suppose that $g_*X \simeq 0$. By compact generation, $X \simeq 0$ if and only if $\mathrm{Hom}(C, X) = 0$ for all $C \in \mathsf{U}^c$. By assumption this is moreover equivalent to $\mathrm{Hom}(g^*C', X) = 0$ for all $C' \in \mathsf{T}^c$. By adjunction, $\mathrm{Hom}(g^*C', X) = \mathrm{Hom}(C', g_*X) = 0$ so $X = 0$ as required. Conversely, suppose that g_* is conservative. To see that $\mathrm{thick}(g^*(\mathsf{T}^c)) = \mathsf{U}^c$ notice that since $g^*(\mathsf{T}^c)$ is a set of compacts (by Lemma 2.23) it suffices by Proposition 2.12 to show that $g^*(\mathsf{T}^c)$ is a set of generators. As such, we need to prove that $\mathrm{Hom}(g^*C, X) = 0$ for all $C \in \mathsf{T}^c$ implies that $X = 0$. Now if $\mathrm{Hom}(g^*C, X) = 0$ for all $C \in \mathsf{T}^c$, then $\mathrm{Hom}(C, g_*X) = 0$ for all $C \in \mathsf{T}^c$ so $g_*X = 0$ by compact generation. Therefore by conservativity $X = 0$ as required. □

Lemma 6.30. *Let $g^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor whose right adjoint g_* is conservative. Then for any $\mathcal{X} \subseteq \mathsf{U}$ we have*

$$g_*(\mathrm{thick}_\otimes(\mathcal{X})) \subseteq \mathrm{thick}_\otimes(g_*\mathcal{X}) \quad \textit{and} \quad g_*(\mathrm{Loc}_\otimes(\mathcal{X})) \subseteq \mathrm{Loc}_\otimes(g_*\mathcal{X}).$$

Proof. Note that $\mathrm{thick}_\otimes(\mathcal{X}) = \mathrm{thick}(\mathcal{X} \otimes \mathsf{U}^c) = \mathrm{thick}(\mathcal{X} \otimes g^*(\mathsf{T}^c))$ where the latter equality holds by Lemma 6.29. As g_* is an exact functor, we therefore have

$$g_* \,\mathrm{thick}(\mathcal{X} \otimes g^*(\mathsf{T}^c)) \subseteq \mathrm{thick}(g_*(\mathcal{X} \otimes g^*(\mathsf{T}^c))) = \mathrm{thick}(g_*\mathcal{X} \otimes \mathsf{T}^c) = \mathrm{thick}_\otimes(g_*\mathcal{X})$$

by the projection formula (Proposition 3.6) as claimed. The statement with localising $\otimes$-ideals follows similarly since g_* is coproduct-preserving. □

Proposition 6.31. *If $f^*\colon \mathsf{T} \to \mathsf{U}$ is an enhanced geometric functor, then $\mathrm{Loc}_\otimes(f^*f_*\mathbb{1}_\mathsf{U}) = \mathsf{U}$.*

Proof. There are maps of commutative algebras

$$\mathbb{1}_\mathsf{U} \xrightarrow{\alpha} f^*f_*\mathbb{1}_\mathsf{U} \xrightarrow{\varepsilon} \mathbb{1}_\mathsf{U}$$

with composite the identity; explicitly, $\alpha = f^*(\eta)$ where $\eta\colon \mathbb{1}_\mathsf{T} \to f_*f^*\mathbb{1}_\mathsf{T} \simeq f_*\mathbb{1}_\mathsf{U}$ is the unit map of the adjunction. The fact that the composite is the identity is then precisely the triangle identity. Write $\varepsilon^*\colon \mathrm{Mod}_\mathsf{U}(f^*f_*\mathbb{1}_\mathsf{U}) \to \mathrm{Mod}_\mathsf{U}(\mathbb{1}_\mathsf{U}) = \mathsf{U}$ for the extension of scalars functor, so that ε_* is restriction of scalars. Therefore, $\varepsilon_*\mathbb{1}_\mathsf{U}$ denotes the underlying object $\mathbb{1}_\mathsf{U}$ equipped with a $f^*f_*\mathbb{1}_\mathsf{U}$-module structure via ε. In particular, $\varepsilon_*\mathbb{1}_\mathsf{U} \in \mathrm{Loc}_\otimes(f^*f_*\mathbb{1}_\mathsf{U})$ in the category $\mathrm{Mod}_\mathsf{U}(f^*f_*\mathbb{1}_\mathcal{D})$ by Proposition 6.28. By applying Lemma 6.30, we deduce that $\mathbb{1}_\mathsf{U} = \alpha_*\varepsilon_*\mathbb{1}_\mathsf{U} \in \mathrm{Loc}_\otimes(\alpha_*f^*f_*\mathbb{1}_\mathsf{U})$ in U. Now it just remains to note that $\alpha_*f^*f_*\mathbb{1}_\mathsf{U}$ is just $f^*f_*\mathbb{1}_\mathsf{U}$ viewed as an object in U (rather than in the category of $f^*f_*\mathbb{1}_\mathsf{U}$-modules). Therefore we have proved the claim. □

Corollary 6.32. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be an enhanced geometric functor. Then f^* is proxy-small if and only if $f_*\mathbb{1}_\mathsf{U}$ is proxy-small in T.*

Proof. Combine Lemma 6.23 with Proposition 6.31. □

7. Grothendieck duality via proxy-smallness

In this section we come to the promised statement that under the proxy-smallness assumption, Grothendieck duality holds on a certain canonical subcategory.

7.A. Classifying rigid objects in the torsion category.

In this section we give various equivalent characterisations of the subcategory of interest.

Definition 7.1. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. We say that $X \in \mathsf{T}$ is *f^*-rigid* if f^*X is rigid in U.

Note that any rigid object is f^*-rigid by Lemma 2.23. However, there are many many more f^*-rigid objects in general:

Example 7.2. For any geometric functor $f^*\colon \mathsf{T} \to \mathsf{U}$ and compact $C \in \mathsf{T}^c$, the object $\Gamma_f C$ is f^*-rigid: we have $f^*\Gamma_f C \simeq f^*C$ by Proposition 6.25. For example, when $f^* = k \otimes^\mathsf{L}_R -$ for a commutative Noetherian local ring $(R, \mathfrak{m}, k)$, the stable Koszul complex $K_\infty(\mathfrak{m}) = \Gamma_f R$ is f^*-rigid, but is not rigid in general.

In order to state and prove the main theorem of this section we set some notation.

Remark 7.3. Let F be a strong symmetric monoidal functor.

(1) Consider the natural map

$$\alpha_{M,N}\colon F\underline{\mathrm{Hom}}(M,N) \to \underline{\mathrm{Hom}}(FM,FN)$$

constructed as the adjunct of

$$FM \otimes F\underline{\mathrm{Hom}}(M,N) \simeq F(M \otimes \underline{\mathrm{Hom}}(M,N)) \xrightarrow{F(\mathrm{ev}_{M,N})} FN$$

This is an isomorphism provided either M or N is rigid. In particular, if M is rigid, then

$$\alpha_{M,\mathbb{1}}\colon F(DM) \xrightarrow{\sim} D(FM)$$

is a natural isomorphism. (Indeed, we already saw this last claim as Lemma 2.23.)

(2) There is a natural transformation

$$\rho_{M,N}\colon DM \otimes N \to \underline{\mathrm{Hom}}(M,N)$$

constructed as the adjunct of

$$M \otimes DM \otimes N \xrightarrow{\mathrm{ev}_{M,\mathbb{1}} \otimes N} \mathbb{1}_{\mathcal{C}} \otimes N \simeq N.$$

(3) There is also a natural transformation

$$\nu_{M,X,N}\colon \underline{\mathrm{Hom}}(M,X) \otimes N \to \underline{\mathrm{Hom}}(M, X \otimes N)$$

constructed as the adjunct of

$$\mathrm{ev}_{M,X} \otimes N\colon M \otimes \underline{\mathrm{Hom}}(M,X) \otimes N \to X \otimes N.$$

Note that $\rho_{M,N} = \nu_{M,\mathbb{1},N}$.

Theorem 7.4. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor. Let $M \in \mathsf{T}$. The following are equivalent:*

(1) the map $\rho_{M,N}\colon DM \otimes N \to \underline{\mathrm{Hom}}(M,N)$ is an f^*-isomorphism for all $N \in \mathsf{T}$, i.e., $f^*(\rho_{M,N})$ is an isomorphism for all $N \in \mathsf{T}$;
(2) $\Gamma_f M \in \Gamma_f \mathsf{T}$ is rigid;
(3) $\Lambda_f M \in \Lambda_f \mathsf{T}$ is rigid;
(4) $f^* M \in \mathsf{U}$ is rigid.

Moreover, these are implied by $\Gamma_f M \in \mathrm{thick}(\Gamma_f(\mathsf{T}^c))$.

Proof. We first prove (1) $\Leftrightarrow$ (3). Write D_{Λ_f} for the functional dual in $\Lambda_f \mathsf{T}$. Note that $\Lambda_f\colon \mathsf{T} \to \Lambda_f \mathsf{T}$ is a strong symmetric monoidal functor (see Remark 6.19), so that Remark 7.3 applies. We have a diagram

$$\begin{tikzcd}
\Lambda_f(DM) \widehat{\otimes} \Lambda_f N \arrow[r, "\simeq"] \arrow[d, "\alpha_{M,\mathbb{1}} \widehat{\otimes} \Lambda_f N"'] & \Lambda_f(DM \otimes N) \arrow[r, "\Lambda_f(\rho_{M,N})"] & \Lambda_f \underline{\mathrm{Hom}}(M,N) \arrow[d, "\alpha_{M,N}"] \\
D_{\Lambda_f}(\Lambda_f M) \widehat{\otimes} \Lambda_f N \arrow[rr, "\rho_{\Lambda_f M, \Lambda_f N}"'] & & \underline{\mathrm{Hom}}(\Lambda_f M, \Lambda_f N)
\end{tikzcd}$$

for any $N \in \mathsf{T}$. This commutes by a diagram chase (omitted here). Since $\Lambda_f \underline{\mathrm{Hom}}(M,N) \simeq \underline{\mathrm{Hom}}(M, \Lambda_f N)$ by Corollary 6.18, the map $\alpha_{M,N}$ is an isomorphism for all $M, N \in \mathsf{T}$. Therefore $\Lambda_f(\rho_{M,N})$ is an isomorphism if and only if $\rho_{\Lambda_f M, \Lambda_f N}$ is an isomorphism if and only if $\Lambda_f M \in \Lambda_f \mathsf{T}$ is rigid. Now by Proposition 6.26, $\Lambda_f(\rho_{M,N})$ is an isomorphism if and only if $f^*(\rho_{M,N})$ is an isomorphism. This proves that (1) is equivalent to (3). Conditions (2) and (3) are equivalent by the MGM equivalence (see Remark 6.19).

For (2) $\Rightarrow$ (4), we have that $f^*\colon \mathsf{T} \to \mathsf{U}$ factorises as

$$\mathsf{T} \xrightarrow{\Gamma_f} \Gamma_f \mathsf{T} \xrightarrow{f^*} \mathsf{U}$$

by Proposition 6.25. Since $f^*\colon \Gamma_f\mathsf{T} \to \mathsf{U}$ is strong symmetric monoidal, and $\Gamma_f M$ is rigid by assumption, it follows that f^*M is rigid, as strong monoidal functors preserve rigid objects, see Lemma 2.23.

To conclude the proof, we prove that (4) $\Rightarrow$ (1). Recall the natural transformation

$$\nu_{M,X,N}\colon \underline{\mathrm{Hom}}(M,X)\otimes N \to \underline{\mathrm{Hom}}(M, X\otimes N)$$

as defined in Remark 7.3, and let $\mathcal{X}$ be the subcategory of $X \in \mathsf{T}$ such that $\nu_{M,X,N}$ is an equivalence for all $N \in \mathsf{T}$. Note that for $C \in \mathcal{C}$ rigid, by naturality $\nu_{M,X\otimes C,N}$ is identified under the obvious isomorphisms with $C \otimes \nu_{M,X,N}$. Thus we see that $\mathcal{X}$ is a thick $\otimes$-ideal of T.

We show that $f_*Y \in \mathcal{X}$ for all $Y \in \mathsf{U}$. To do so, we note that we have isomorphisms fitting into a diagram

$$\begin{tikzcd}
\underline{\mathrm{Hom}}(M, f_*Y)\otimes N \arrow[r, "\nu_{M,f_*Y,N}"] \arrow[d, "\simeq"', "(3.8)"] & \underline{\mathrm{Hom}}(M, f_*Y\otimes N) \arrow[d, "(3.6)"', "\simeq"] \\
f_*\underline{\mathrm{Hom}}(f^*M, Y)\otimes N \arrow[d, "\simeq"', "(3.6)"] & \underline{\mathrm{Hom}}(M, f_*(Y\otimes f^*N) \arrow[d, "(3.8)"', "\simeq"] \\
f_*(\underline{\mathrm{Hom}}(f^*M,Y)\otimes f^*N) \arrow[r, "f_*(\nu_{f^*M,Y,f^*N})"'] & f_*(\underline{\mathrm{Hom}}(f^*M, Y\otimes f^*N)
\end{tikzcd}$$

commutes. Since f^*M is rigid, ν_{f^*M,Y,f^*N} is an isomorphism for all N, and therefore by the above diagram we see that $\nu_{M,f_*Y,N}$ is also an isomorphism so $f_*Y \in \mathcal{X}$ as claimed.

Since $\mathcal{X}$ is a thick $\otimes$-ideal, it follows that $\mathcal{W} \subseteq \mathcal{X}$ where $\mathcal{W}$ is a witness for proxy-smallness as $\mathcal{W} \subseteq \mathrm{thick}_\otimes(f_*\mathbb{1}_\mathsf{T})$. Since each $W \in \mathcal{W}$ is rigid, we have that $\nu_{M,W,Y}$ corresponds under natural isomorphisms to $W \otimes \nu_{M,\mathbb{1}_\mathsf{T},Y}$. As $\nu_{M,\mathbb{1}_\mathsf{T},Y} = \rho_{M,Y}$, this proves that $W \otimes \rho_{M,Y}$ is an isomorphism for all $W \in \mathcal{W}$ and $Y \in \mathsf{T}$, which by Proposition 6.26 gives (1).

For the final statement, since $\Gamma_f\colon \mathsf{T} \to \Gamma_f\mathsf{T}$ is strong monoidal it preserves rigid objects (see Lemma 2.23). Therefore any object in $\Gamma_f(\mathsf{T}^c)$ is rigid in $\Gamma_f\mathcal{C}$, so the claim follows by a thick subcategory argument. □

Remark 7.5. We note that the condition that $\Gamma_f M \in \mathrm{thick}(\Gamma_f(\mathsf{T}^c))$ is not equivalent to the other conditions in the previous theorem in general. The counterexample comes from the map of ring spectra $L_nS^0 \to K(n)$ and is way beyond the scope of this course.

7.B. **Case study: local algebra.** We now give an explicit example of the previous theorem in action. For a commutative Noetherian local ring $(R, \mathfrak{m}, k)$, we consider the geometric functor $f^* = k \otimes^{\mathsf{L}}_R -\colon \mathsf{D}(R) \to \mathsf{D}(k)$ and write $\Gamma_\mathfrak{m} = \Gamma_f$ and similarly for $\Lambda_\mathfrak{m}$.

Remark 7.6. The functor $\Gamma_\mathfrak{m}$ computes the total right derived functor of the $\mathfrak{m}$-power torsion functor, and the functor $\Lambda_\mathfrak{m}$ computes the total left derived functor of the $\mathfrak{m}$-adic completion functor.

In this case, we can in fact show that slightly more than Theorem 7.4 claims is true. In order to do so we need the following two lemmas:

Lemma 7.7. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring, and let $M \in \Lambda_\mathfrak{m}\mathsf{D}(R)$. If $H_*(k \otimes^{\mathsf{L}}_R M)$ is bounded, then the natural map*

$$M \simeq R \otimes^{\mathsf{L}}_R M \to k \otimes^{\mathsf{L}}_R M$$

is non-zero in homology.

Proof. To prove this we need the following "black box":

If $N \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$, then $H_n(N)/\mathfrak{m}H_n(N) = 0$ implies that $H_n(N) = 0$.

The proof of the above black box is not too difficult but it would take us too far from the point so we omit it. Since $H_*(k \otimes^{\mathsf{L}}_R M)$ is bounded by assumption, it follows that $H_*(K(\mathfrak{m}) \otimes^{\mathsf{L}}_R M)$ is also bounded by Theorem 5.16.

Claim: Let $x \in R$ and $X \in \mathsf{D}(R)$. If $H_*(K(x) \otimes_R X)$ is bounded, then H_*X is bounded.

This claim holds since the assumption means that x acts as an isomorphism on $H_n(X)$ for $n \ll 0$. Therefore by the black box, we have $H_n(X) = 0$.

Given this claim, by induction we deduce from $H_*(K(\mathfrak{m}) \otimes_R M)$ being bounded, that $H_*(M)$ is bounded. Now, write $i = \inf\{n \mid H_n(M) \neq 0\}$. One checks that $H_i(k \otimes^{\mathsf{L}}_R M) = k \otimes_R H_iM = H_iM/\mathfrak{m}H_iM$. Therefore the canonical map induces the non-zero quotient map

$$H_iM \to H_i(k \otimes^{\mathsf{L}}_R M) = H_iM/\mathfrak{m}H_iM$$

in homology. □

Lemma 7.8. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring and let $M \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$. If $\operatorname{rank}_k H_*(k \otimes^{\mathsf{L}}_R M)$ is finite then $M \in \operatorname{thick}(\Lambda_{\mathfrak{m}}R)$.*

Proof. We argue by induction on $r = \operatorname{rank}_k(H_*(k \otimes^{\mathsf{L}}_R M))$. For the base case $r = 0$, this means that $k \otimes^{\mathsf{L}}_R M \simeq 0$. Therefore by Proposition 6.26 we see that $M \simeq \Lambda_{\mathfrak{m}}M \simeq 0$ giving the base case.

Now suppose that $r \geq 1$. Since the rank is finite, we have that $H_*(k \otimes^{\mathsf{L}}_R M)$ is bounded. Therefore by Lemma 7.7, we have a non-zero map $\alpha\colon H_nM \to H_n(k \otimes^{\mathsf{L}}_R M)$ for some n. Since $H_n(-) = \operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^n R, -)$, this non-zero map α is equivalent to the data of a map $\Sigma_n R \to M$ such that the induced map $\Sigma^n k \to k \otimes^{\mathsf{L}}_R M$ is non-zero.

The map α gives rise to a map $\Sigma^n \Lambda_{\mathfrak{m}} R \to M$ since $M \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$, and therefore to a triangle

$$\Sigma^n \Lambda_{\mathfrak{m}} R \to M \to N.$$

Applying $k \otimes^{\mathsf{L}}_R -$ we get a triangle

$$\Sigma^n k \to k \otimes^{\mathsf{L}}_R M \to k \otimes^{\mathsf{L}}_R N$$

in $\mathsf{D}(k)$ in which the first map is non-zero. Therefore taking the long exact sequence in homology, we see that $\operatorname{rank}_k(H_*(k \otimes^{\mathsf{L}}_R M)) = \operatorname{rank}_k(H_*(k \otimes^{\mathsf{L}}_R N)) + 1$. Therefore by inductive hypothesis we have that $N \in \operatorname{thick}(\Lambda_{\mathfrak{m}}R)$. Hence by the above triangle, we also have $M \in \operatorname{thick}(\Lambda_{\mathfrak{m}}R)$. □

As an aside, we note a corollary of the proof of the previous lemma which will become relevant later on in the course:

Corollary 7.9. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring and let $M \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$. If $k \otimes^{\mathsf{L}}_R M \simeq \Sigma^a k$ for some $a \in \mathbb{Z}$, then $M \simeq \Sigma^a \Lambda_{\mathfrak{m}} R$.*

Proof. The assumption that $k \otimes^{\mathsf{L}}_R M \simeq \Sigma^a k$ says that $\operatorname{rank}_k H_*(k \otimes^{\mathsf{L}}_R M) = 1$. The proof of the inductive step in Lemma 7.8 then gives the claim. □

Proposition 7.10. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring. The following are equivalent for $M \in \mathsf{D}(R)$:*

(1) $\Gamma_{\mathfrak{m}}M \in \Gamma_{\mathfrak{m}}\mathsf{D}(R)$ is rigid;
(2) $k \otimes^{\mathsf{L}}_R M \in \mathsf{D}(k)$ is rigid;

(3) $\operatorname{rank}_k H^*(k \otimes^{\mathsf{L}}_R M)$ *is finite;*
(4) $\Gamma_{\mathfrak{m}} M \in \operatorname{thick}(\Gamma_{\mathfrak{m}} R)$.

Proof. Conditions (1) and (2) are equivalent and are implied by (4) by Theorem 7.4. Condition (3) is a rephrasing of condition (2) since k is a field. So it suffices to prove that (3) implies (4). As $k \simeq \Gamma_{\mathfrak{m}} k$, we see that the canonical map

$$k \otimes^{\mathsf{L}}_R X \to k \otimes^{\mathsf{L}}_R \Lambda_{\mathfrak{m}} X$$

is an isomorphism, by the MGM equivalence and the fact that $\Gamma_{\mathfrak{m}}$ is smashing. Therefore by Lemma 7.8 we see that $\Lambda_{\mathfrak{m}} X \in \operatorname{thick}(\Lambda_{\mathfrak{m}} R)$. Now the MGM equivalence (see Theorem 6.15) restricts to an equivalence of categories

$$\operatorname{thick}(\Gamma_{\mathfrak{m}} R) \underset{\Lambda_{\mathfrak{m}}}{\overset{\Gamma_{\mathfrak{m}}}{\underset{\longrightarrow}{\overset{\longleftarrow}{\simeq}}}} \operatorname{thick}(\Lambda_{\mathfrak{m}} R)$$

from which we deduce that $\Gamma_{\mathfrak{m}} X \simeq \Gamma_{\mathfrak{m}} \Lambda_{\mathfrak{m}} X \in \operatorname{thick}(\Gamma_{\mathfrak{m}} R)$ as required. □

7.C. **Grothendieck duality.** We now move onto the consequences of Theorem 7.4, and show that f^*-rigidity of $M \in \mathsf{T}$ is usually sufficient for M to enjoy properties in relation to f^* that one would typically only expect if M itself were rigid. These properties will allow us to prove Grothendieck duality for f^*-rigid objects below.

Corollary 7.11. *Suppose* $f^* \colon \mathsf{T} \to \mathsf{U}$ *is a proxy-small geometric functor and let* $M \in \mathsf{T}$. *If* $f^* M$ *is rigid, then the canonical map*

$$f^*(d_M) \colon f^* M \to f^*(D^2 M)$$

is an isomorphism.

Proof. By Proposition 6.26 it suffices to prove that $\Lambda_f(d_M)$ is an isomorphism. Since $f^* M$ is rigid in U, by Theorem 7.4 we have that $\Lambda_f M$ is rigid in $\Lambda_f \mathsf{T}$ and hence the natural map $d'_{\Lambda_f M} \colon \Lambda_f M \to D^2_{\Lambda_f}(\Lambda_f M)$ is an isomorphism, where D_{Λ_f} denotes the functional dual in $\Lambda_f \mathsf{T}$. Recall from Proposition 6.12 that the internal hom in complete modules is computed in the ambient category and that the tensor unit is $\Lambda_f \mathbb{1}$. Since $\Lambda_f \colon \mathsf{T} \to \Lambda_f \mathsf{T}$ is closed monoidal, there is a natural isomorphism $D^2_{\Lambda_f}(\Lambda_f(-)) \simeq \Lambda_f D^2(-)$ making the diagram

$$\begin{tikzcd} \Lambda_f M \arrow[r, "\Lambda_f(d_M)"] \arrow[dr, "d'_{\Lambda_f M}"', "\simeq"] & \Lambda_f D^2 M \arrow[d, "\simeq"] \\ & D^2_{\Lambda_f}(\Lambda_f M) \end{tikzcd}$$

commute. We see from this diagram that $\Lambda_f(d_M)$ is an isomorphism as claimed. □

Corollary 7.12. *Suppose* $f^* \colon \mathsf{T} \to \mathsf{U}$ *is a proxy-small geometric functor and let* $M \in \mathsf{T}$. *Recall the natural map* $\alpha_{M,N} \colon f^* \underline{\operatorname{Hom}}(M, N) \to \underline{\operatorname{Hom}}(f^* M, f^* N)$ *from Remark 7.3. If* $f^* M$ *is rigid, then for all* $N \in \mathsf{T}$ *there are natural isomorphisms:*

$$\alpha_{M,N} \colon f^* \underline{\operatorname{Hom}}(M, N) \xrightarrow{\sim} \underline{\operatorname{Hom}}(f^* M, f^* N) \tag{7.13}$$

$$\alpha_{N,M} \colon f^* \underline{\operatorname{Hom}}(N, M) \xrightarrow{\sim} \underline{\operatorname{Hom}}(f^* N, f^* M) \tag{7.14}$$

$$\alpha_{M,\mathbb{1}} \colon f^*(DM) \xrightarrow{\sim} D(f^* M). \tag{7.15}$$

Proof. Recall from Proposition 6.25 that $f^*\colon \mathsf{T} \to \mathsf{U}$ admits a factorisation

$$\mathsf{T} \xrightarrow{\Gamma_f} \Gamma_f\mathsf{T} \xrightarrow{f_t^*} \mathcal{D}$$

where f_t^* is the restriction of f^* to $\Gamma_f\mathsf{T}$. We note that f_t^* is also strong symmetric monoidal. Thus, the construction of Remark 7.3 composes, i.e., $\alpha_{M,N}$ is the composition

$$f_t^*(\Gamma_f\underline{\mathrm{Hom}}_{\mathsf{T}}(M,N)) \xrightarrow{f_t^*(\gamma_{M,N})} f_t^*\underline{\mathrm{Hom}}_{\Gamma_f\mathsf{T}}(\Gamma_f M, \Gamma_f N) \xrightarrow{\beta_{\Gamma_f M,\Gamma_f N}} \underline{\mathrm{Hom}}_{\mathsf{U}}(f_t^*\Gamma_f M, f_t^*\Gamma_f N)$$

where β and γ are the natural transformations analogous to α associated to f_t^* and Γ_f. Since $\Gamma_f\colon \mathsf{T} \to \Gamma_f\mathsf{T}$ is closed monoidal, γ is a natural isomorphism, and hence $f_t^*(\gamma_{M,N})$ is an isomorphism. Therefore, to prove (7.13) it suffices to prove that $\beta_{\Gamma_f M,\Gamma_f N}$ is an isomorphism. Since f^*M being rigid implies $\Gamma_f M \in \Gamma_f\mathsf{T}$ is rigid by Theorem 7.4, this is immediate from Remark 7.3. The proof of (7.14) is analogous, noting that $\beta_{\Gamma_f N,\Gamma_f M}$ is also an isomorphism since $\Gamma_f M$ is rigid. Finally, (7.15) is the case $N = \mathbb{1}_{\mathcal{C}}$ in (7.13). □

We are now ready to prove the promised generalisation of Grothendieck duality. Recall from Lemma 4.3 that rigid objects have Grothendieck duality. The following shows that under the proxy-small hypothesis, this extends to the larger class of f^*-rigid objects:

Theorem 7.16. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor, and $M \in \mathsf{T}$. If f^*M is rigid, then there is a natural isomorphism*

$$f^!X \otimes f^*M \simeq f^!(X \otimes M) \tag{7.17}$$

for all $X \in \mathsf{T}$. In particular, if f^*M is rigid there is an isomorphism

$$\omega_f \otimes f^*M \simeq f^!M. \tag{7.18}$$

Proof. We have natural isomorphisms

$$f^!X \otimes f^*M \simeq \underline{\mathrm{Hom}}(D(f^*M), f^!X) \simeq \underline{\mathrm{Hom}}(f^*(DM), f^!X) \simeq f^!\underline{\mathrm{Hom}}(DM, X)$$

using the rigidity of f^*M, (7.15), and then (3.10). To conclude it therefore suffices to show that the canonical map

$$M \otimes X \xrightarrow{d_M \otimes X} D^2M \otimes X \xrightarrow{\rho_{DM,N}} \underline{\mathrm{Hom}}(DM, X)$$

is an $f^!$-isomorphism, where $\rho_{DM,N}$ is as in Remark 7.3. By Proposition 6.26, this is equivalent to checking that it is an f^*-isomorphism.

Note that since f^*M is rigid, it follows that $f^*(DM)$ is also rigid (since it is isomorphic to $D(f^*M)$ by (7.15)). Now consider the following diagram in which the left hand vertical is the map we want to show is an isomorphism:

$$\begin{tikzcd}
f^*(M \otimes X) \arrow[r, "\simeq"] \arrow[d, "f^*(d_M \otimes X)"'] & f^*M \otimes f^*X \arrow[d, "f^*(d_M)\otimes f^*X", "\simeq\ (7.11)"'] & \\
f^*(D^2M \otimes X) \arrow[r, "\simeq"] \arrow[d, "f^*(\rho_{DM,N})"'] & f^*(D^2M) \otimes f^*X \arrow[r, "\alpha_{DM,\mathbb{1}}\otimes f^*X", "\simeq\ (7.15)"'] & D(f^*DM) \otimes f^*X \arrow[d, "\rho_{f^*DM,f^*X}", "\simeq"'] \\
f^*\underline{\mathrm{Hom}}(DM, X) \arrow[rr, "\simeq\ (7.13)", "\alpha_{DM,X}"'] & & \underline{\mathrm{Hom}}(f^*DM, f^*X)
\end{tikzcd}$$

Note that the diagram commutes: the top square by the strong monoidality of f^*, and the bottom rectangle by a diagram chase left to the reader. Therefore, we see that the claimed map is an isomorphism as required. This completes the proof of (7.17). Taking $X = \mathbb{1}_{\mathsf{T}}$ in (7.17) gives (7.18). $\square$

8. The Wirthmüller isomorphism

9. Matlis duality

10. Orientability and Gorenstein duality

Appendix A. Exercises

Exercise A.1. Prove Proposition 2.11.

Exercise A.2. Prove that for any ring R, the compact objects in $\mathsf{D}(R)$ are the perfect complexes.

Exercise A.3. Prove that if $M \in \mathsf{stmod}kG$ (that is, M is finitely generated), then M is compact in $\mathsf{StMod}(kG)$.

Exercise A.4. Let T be a tt-category and let $Y \in \mathsf{T}$ be rigid. Prove that the natural map

$$F(X, \mathbb{1}) \otimes Y \to F(X, Y)$$

is an isomorphism for all $X \in \mathsf{T}$.

Exercise A.5. Let T be a tensor-triangulated category and $X \in \mathsf{T}$ be rigid. Prove that if Y is internally compact (resp., rigid), then $X \otimes Y$ is internally compact (resp., rigid). (See Proposition 2.31.)

Exercise A.6. Let $F\colon \mathsf{T} \to \mathsf{U}$ be a tt-functor between tt-categories. Let $X \in \mathsf{T}$ be rigid, and $Y \in \mathsf{T}$ be arbitrary. Prove that there is a natural isomorphism $F\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(FX, FY)$.

Exercise A.7. Let $\mathcal{K}$ be a set of compact objects in a rigidly-compactly generated tt-category T. Consider the localising $\otimes$-ideal $\mathsf{L} = \mathrm{Loc}_{\otimes}(\mathcal{K})$. Explain how one can view this as a tt-category (note that it is *not* a tt-subcategory of T). Which of the following are true in general?:

(1) if X is compact in L then X is compact in T;
(2) if $X \in \mathsf{L}$ is compact in T then X is compact in L;
(3) if X is rigid in L then X is rigid in T;
(4) if $X \in \mathsf{L}$ is rigid in T then X is rigid in L;
(5) $\{\text{compacts in } \mathsf{L}\} \subset \{\text{rigids in } \mathsf{L}\}$;
(6) $\{\text{compacts in } \mathsf{L}\} \supset \{\text{rigids in } \mathsf{L}\}$;
(7) $\{\text{compacts in } \mathsf{L}\} = \{\text{rigids in } \mathsf{L}\}$.

Exercise A.8. Let T be a tt-category and $X \in \mathsf{T}$ be rigid. Prove that X is a retract of $X \otimes DX \otimes X$. Hence deduce that for any rigid X we have $\mathrm{thick}_{\otimes}(X) = \mathrm{thick}_{\otimes}(DX)$.

Exercise A.9. Prove Lemma 3.1.

Exercise A.10. Prove implication (5) $\Rightarrow$ (2) in Theorem 4.8. (Hint: use Lemma 4.6.)

Exercise A.11. Which of the isomorphisms listed in Corollary 4.10 imply Grothendieck–Neeman duality?

Exercise A.12. Which of the following are true? Give proofs or counterexamples accordingly:

(1) If X is proxy-small, then ΣX is proxy-small.
(2) If $\{X_i\}_{i \in I}$ is a set of proxy-small objects, then $\oplus X_i$ is proxy-small.
(3) $\mathbb{Q}$ is proxy-small in $\mathsf{D}(\mathbb{Z})$.
(4) Let R be a commutative ring, and $M \in \mathsf{D}(R)$ be arbitrary. Then $R \oplus M$ is proxy-small in $\mathsf{D}(R)$.
(5) If $X \oplus Y$ is proxy-small, then X and Y are proxy-small.
(6) Let $f\colon X \to Y$ be a map between proxy-small objects. Complete it to a triangle $X \to Y \to Z$. Then Z is proxy-small.

Exercise A.13. Let T be a tensor-triangulated category. Prove that the one-to-one correspondence between localizations and colocalizations in Proposition 6.5 restricts to a one-to-one correspondence between monoidal localizations and monoidal colocalizations.

Exercise A.14. Let T be a tensor-triangulated category, and L be a monoidal localization. Prove that the natural map $\alpha_X \colon L\mathbb{1} \otimes X \to LX$ from Lemma 6.10 is an isomorphism if X is rigid.

Exercise A.15. Let T be a tensor-triangulated category, L be a monoidal localization, and write $\mathcal{S}$ for the full subcategory of L-local objects.

(1) Prove Proposition 6.11.
(2) Give an example of a smashing localization.
(3) Prove that $L \colon \mathsf{T} \to \mathcal{S}$ preserve compacts if L is smashing. Does this still hold if L is not smashing?

Exercise A.16. Let T be a rigidly-compactly generated tt-category and $L \colon \mathsf{T} \to \mathsf{T}$ be a monoidal localisation functor. Prove that $L\mathsf{T}$ is rigidly-compactly generated if and only if L is smashing.

Exercise A.17. Let $(\mathsf{T}, \mathcal{K})$ be a local duality context, and assume that $\mathcal{K} = \{K\}$ is a singleton. An object $X \in \mathsf{T}$ is said to be *homologically K-local* if $\mathrm{Hom}(Y, X) \simeq 0$ whenever $K \otimes Y \simeq 0$. A map $f \colon X \to Y$ in T is said to be a *K-equivalence* if $K \otimes f$ is an equivalence. This question is about showing that completion may be equivalently described as homological K-localization.

(1) Show that X is homologically K-local if and only if for all $f \colon Y \to Z$ such that $K \otimes f$ is an equivalence, the induced map $\mathrm{Hom}(f, X)$ is an equivalence.
(2) Prove that the map $X \to \Lambda X$ is a K-equivalence.
(3) Prove that X is complete if and only if X is homologically K-local.